

Faculty of Informatics Engineering
Department of Software Engineering

Traffic Police Assistance System

A senior 1 project report- Completed the requirements for obtaining a bachelor's degree in Informatics Engineering - Software Engineering and Information Systems

Prepared by

Tasnim aldakak

maryam albatool almasri

Supervised by

Dr. Riad Sonbol

Eng. Reghad alhosny

January 2026

ABSTRACT

This project addresses the design and implementation of an **Intelligent Traffic Police Assistant System**, aimed at providing traffic officers with a smart and integrated technological platform to support their daily operational tasks with greater efficiency and accuracy. Classified within the domain of **intelligent information systems and artificial intelligence applications in traffic management**, the project contributes to improving traffic violation handling processes and enhancing overall road safety.

The motivation behind this project arises from the continued reliance on traditional and manual procedures for recording traffic violations, which often lead to operational delays, susceptibility to human error, and difficulties in managing and analyzing violation data. Furthermore, many existing systems lack effective integration with modern artificial intelligence technologies, such as optical character recognition and speech processing, limiting their usability in real-world, field-based environments.

The primary objective of the project is to develop a reliable and intelligent system that enables traffic police officers to efficiently record traffic violations through **automatic vehicle data extraction from images using Optical Character Recognition (OCR)**, as well as **voice-based reporting through Speech-to-Text (STT) technologies**. The system also facilitates the structured management of traffic records, violation logs, and the generation of analytical reports to support monitoring and decision-making processes.

The project follows an analytical and applied methodology, beginning with a comprehensive analysis of system and user requirements, followed by the design of a normalized relational database using MySQL. Backend services are implemented using the Laravel framework, adopting a **microservices-based and message-driven architecture** with RabbitMQ to ensure scalability and loose coupling. The client-side application is developed using Flutter to deliver a responsive and user-friendly interface suitable for mobile devices. Testing and validation were conducted using tools such as Postman to ensure the correctness and reliability of the system's APIs, while Firebase was optionally integrated to support real-time notification functionality.

The final outcome is a coherent, intelligent, and user-centered traffic police assistant system that enhances operational efficiency, reduces reporting time, improves data accuracy, and supports traffic officers in enforcing regulations more effectively, thereby contributing to safer and more organized road traffic management.

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List of abbreviations

Abbreviation	Full Term
API	Application Programming Interface
DB	Database
DBMS	Database Management System
SQL	Structured Query Language
JWT	JSON Web Token
CRUD	Create, Read, Update, Delete
OCR	Optical Character Recognition
STT	Speech-to-Text
AI	Artificial Intelligence
UI	User Interface
HTTP	HyperText Transfer Protocol
URL	Uniform Resource Locator
JSON	JavaScript Object Notation
OOP	Object-Oriented Programming
RMQ	RabbitMQ Message Broker

Chapter 1 Introduction

Introduction:

In this chapter, we will introduce our project, discussing the main issues and reasons for building this system. We will also explain the objectives and goals we aim to accomplish with this system. Finally, we will provide an overview the report organization.

Problem Definition:

In the field of traffic management and law enforcement, traffic police officers are responsible for maintaining road safety and enforcing regulations. However, many traffic monitoring and violation reporting processes still rely on manual methods, which are often inefficient, time-consuming, and prone to human error.

Traditional traffic violation systems commonly lack real-time processing, intelligent automation, and effective analytical capabilities, making it difficult to accurately manage and analyze large volumes of violation data. These limitations reduce operational efficiency and hinder effective field support for officers.

This project addresses these challenges by developing a **Traffic Police Assistant System** that enables officers to efficiently record traffic violations using **OCR** for automatic vehicle data extraction and **STT** for converting voice notes into structured text. The system also supports organized violation management and report generation.

The system is developed using modern technologies, including Laravel for backend services, Flutter for the mobile interface, and MySQL for database management, following a service-oriented and microservices-based architecture. Testing tools such as Postman are used to ensure reliability, with optional Firebase integration for notifications.

By modernizing traffic violation reporting, the proposed system improves efficiency, reduces errors, and enhances data accuracy, ultimately supporting effective traffic law enforcement and improved road safety.

Project objectives:

The primary objective of this project is to develop an **Traffic Police Assistant System** that supports traffic police officers in efficiently recording, managing, and monitoring traffic violations, while ensuring that the system's functionalities align with real-world field operations to enhance accuracy, usability, and overall performance.

To achieve this objective, the system provides a set of intelligent features that enable officers to capture vehicle images and automatically extract relevant information using **Optical Character Recognition (OCR)** technologies, as well as record verbal observations that are converted into structured textual data

through **Speech-to-Text (STT)** mechanisms. The system also facilitates the organization and classification of traffic violations, enabling efficient record management and retrieval.

In addition, the system generates structured reports and analytical summaries that assist officers and administrators in reviewing violation data, identifying traffic patterns, and supporting informed decision-making. These analytical tools help improve situational awareness and provide insights into recurring violations and high-risk areas.

Ultimately, the project aims to simplify and modernize traffic violation management by providing a reliable, intelligent, and data-driven platform that enhances operational efficiency, reduces human error, and contributes to improved traffic law enforcement and road safety.

Report organization:

The report will be organized in the following chapters:

- Chapter 1: introduction.
- Chapter 2: Basic concepts and literature review.
- Chapter 3: project management.
- Chapter 4: System analysis.
- Chapter 5: System Design
- Chapter 6: practical implementation.
- Chapter 7: Conclusion.

Chapter 2 Project Management

1. Introduction:

In this chapter, we will dive into the management phase of the project, which is a critical aspect of ensuring the project's success. We will examine the project charter, project plan, Statement of Work (SOW) document, stakeholder analysis, and risk management strategies to effectively guide and control the project from initiation to completion.

2. Project charter:

A project charter is a formal document that serves as an official authorization for the start of a project. It acts as a reference point throughout the project, providing a clear understanding of the project's purpose and establishing a foundation for decision-making and project governance

Project Title	Traffic police assistant
Start Date	october 30, 2025
Finish Date	June 10, 2026
Project Manager(s)	Dr. Riad Sonbol, Eng. Raghad Al Hosny
Project Objectives	Automate violation recording using voice-to-text services, vehicle plate recognition, and automatically generate reports. Providing customized interfaces for each role (police_officer, police_manager, administrator, citizen). Enabling citizens to follow up on violations and submit appeals electronically Unifying data into a single platform that supports transparency and speed of decision-making
Project Approach	The project will follow a step-by-step approach with testing after each phase: Determining the scope and objectives of the project. Performing a literature review of related systems and technologies.

	☐ Analyzing the functional and non-functional requirements of the system. Performing system decomposition and architectural design. Starting the first increment and implementing the authentication and account management services. Testing the functionalities of the first increment. Starting the second increment and implementing the traffic violation management services. Testing the functionalities of the second increment. Starting the third increment and implementing the artificial intelligence service for vehicle plate recognition (OCR). Testing the functionalities of the third increment. Starting the fourth increment and implementing the artificial intelligence service for speech-to-text processing. Testing the functionalities of the fourth increment. Documenting every step of the development process
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Roles and Responsibilities:

Dr. Riad Sonbol	Project Manager / Supervisor	General supervision and project oversight
Raghad Al Hossny	supervisor	Project coordination, progress monitoring, documentation support
Mariam Albatool Almasri	Engineer	Backend development, API implementation , AI Contributor ,system documentation
Tasnim Aldakak	Engineer	Frontend development, UI design, AI Contributor , system documentation

3. The SOW document

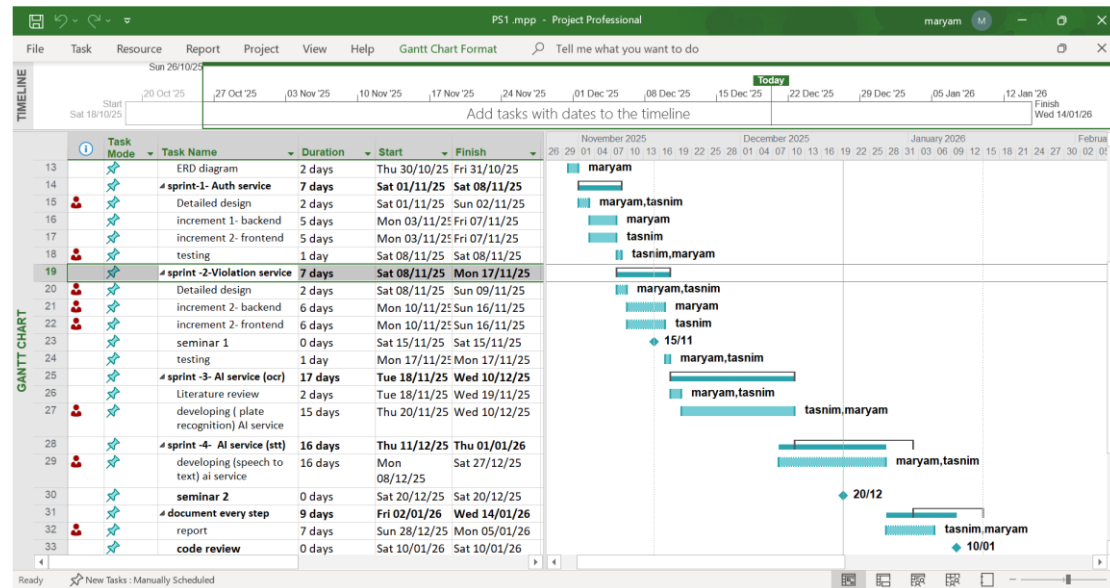
A Statement of Work (SOW) is a comprehensive document that defines the scope of work for a project. It outlines the specific tasks, deliverables, timeline, and responsibilities. The SOW document provides a clear understanding of what needs to be accomplished, the project's objectives, and the criteria for success.

Project Title	Traffic police assistant
Project Description	An intelligent system that assists traffic police officers in recording and managing traffic violations using artificial intelligence technologies. The system supports police officers through a smart mobile application that facilitates violation documentation using AI-based services, provides an administrator responsible for managing user accounts and system settings, enables citizens to view their recorded violations and submit appeals electronically, and allows police managers to review violations and appeals and make approval or rejection decisions
Objectives	Automate violation recording using voice-to-text services, vehicle plate recognition, and automatically generate reports. Providing customized interfaces for each role (police officer, police manager, administrator, citizen). Enabling citizens to follow up on violations and submit appeals electronically Unifying data into a single platform that supports transparency and speed of decision-making
Project Scope	Managing users and roles, including citizens, police officers, police managers, and administrators. Providing artificial intelligence services such as speech-to-text conversion, vehicle license plate recognition, and automatic violation report generation. Organizing and recording traffic violations by police officers through a smart field application. Enabling the Traffic Appeals Department to track appeal status from submission to final decision. Providing role-based control panels to display violations, statistics, and permitted operations. Maintaining a central database for storing violations, reports, and related attachments. Offering a citizen portal for tracking violations, submitting appeals
Project Goals	Automate traffic violation recording and documentation using artificial intelligence technologies.

	Improve the accuracy and efficiency of traffic law enforcement operations. Enable electronic management and tracking of traffic violations and appeals. Provide role-based access and interactive dashboards for different system users. Enhance transparency by allowing citizens to view violations and submit appeals online. Support decision-making through centralized data storage and reporting
Deliverables	Project plan, SRS document, final report, backend and frontend system components, and artificial intelligence components including speech-to-text services, vehicle license plate recognition
Technologies Used	Programming Languages & Frameworks: Python, PHP, Django REST Framework, Laravel, Flutter Databases: MySQL, MongoDB Artificial Intelligence Technologies: Speech-to-Text models, Optical Character Recognition (OCR) models for vehicle plate recognition Tools: vs, GitHub, Postman, RabbitMQ
Assumptions	Team members are continuously available during development. Required AI technologies and development tools are accessible. The development environment remains stable throughout the project
Team Members & Roles	Dr. Riad Sonbol – Project Manager Eng.raghad alhossny – supervisor Tasnim Aldakak – Frontend Developer, AI Contributor Mariam albatool Almasri – Backend Developer, AI Contributor
Schedule	Start: october 30, 2025 First Seminar: november 15, 2025 Second Project Seminar: December 20, 2025 Technical Committee Review: January 11, 2026 Finish: June 10, 2026 Final Project Seminar: January 31, 2026

4. Project plan – Gantt chart:

A document outlining tasks, deadlines, and resources needed to achieve project objectives, serving as a roadmap for successful project execution.



5. Risk Management:

the process of identifying, evaluating, and mitigating potential risks that could impact the success of a project or a team.

Risk_I D	Risk title	Risk Description	Raised Date	Trackin g Freque ncy	State	Impact	Mitigation Plan
RK-01	team of only two students	if one of the students stops for some reason, the whole project progress will be impacted	22/10/2025	Weekly	active	high	the one who end first will help the other one, in his work
RK-02	Learning New Technologies	in our project, we need to use some AI components and our knowledge of developing it where low and advanced database concepts.	12/11/2025	Weekly	under mitigation	medium	Allocate time for learning, use official documentation, tutorials, and seek academic guidance when needed

RK-03	Integration Challenges	Difficulties may arise when integrating backend APIs with the Flutter frontend, web frontend and AI services	30/10/2025	Weekly	active	high	Perform regular integration sprints, continuous API testing using Postman, and maintain clear API documentation
RK-04	Data Security & Privacy	The system uses two databases (MySQL for core system data and MongoDB for AI-related data). Improper separation or security configuration may lead to data inconsistency or privacy risks	20/10/2025	Weekly	active	high	Clearly separate database responsibilities, apply encryption and access control policies, and follow security best practices for both databases
RK-05	R big Separation of work in the implementation stage	Dividing responsibilities between backend, frontend, and artificial intelligence components may lead to coordination challenges during the implementation stage	15/12/2025	daily	active	medium	Assign an implementation leader and conduct regular coordination meetings to ensure alignment between all system components

6. Version Control and Configuration Management:

6.1 Project Description

The *Traffic Police Assistance* project is a multi-service system developed using a microservices-based architecture. The system consists of backend services, frontend applications, and artificial intelligence services that work together to provide intelligent traffic violation management. Due to the complexity of the system and the involvement of multiple technologies and components, version control and configuration management were essential to ensure organized development, controlled integration, and system stability throughout the project lifecycle.

6.2 Git Repository Structure

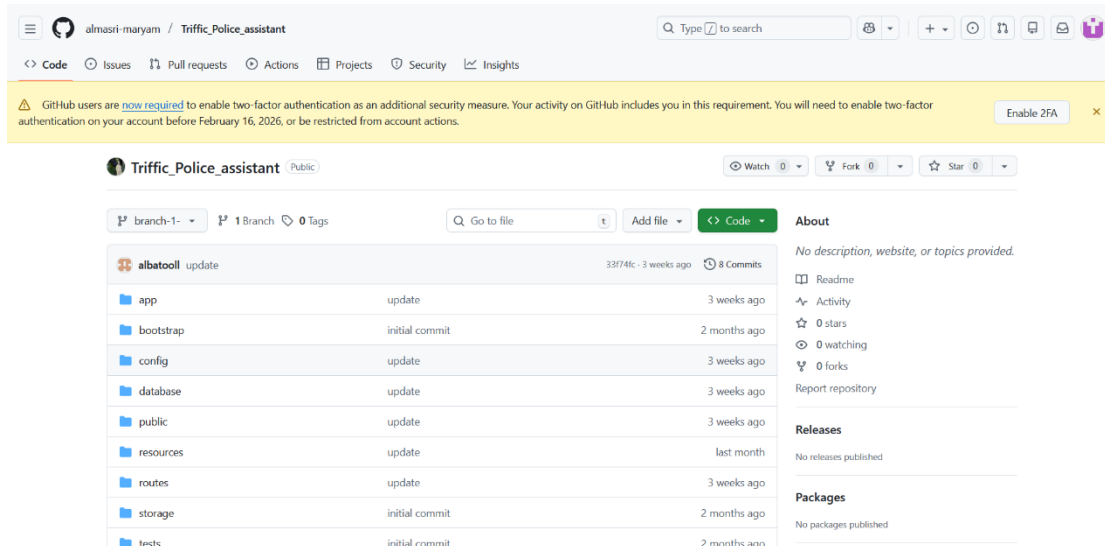
Version control for the project is managed using Git and hosted on GitHub. The backend source code is maintained in a dedicated repository that contains all backend microservices.

Backend Repository:

- **Repository Name:** `Triffic_police_assistant`
- **Purpose:** Contains the backend source code responsible for core system logic, user management, violation management, and integration with AI services.

The repository is structured to keep each microservice isolated, enabling independent development and deployment.

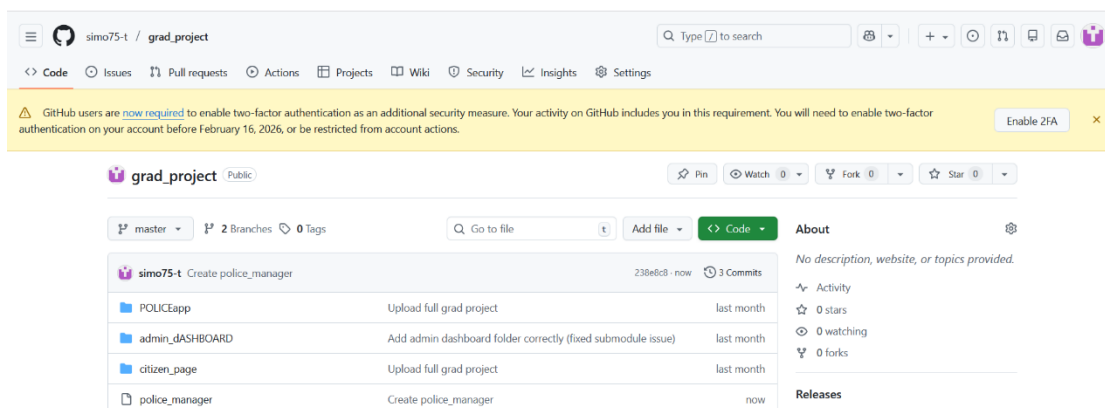
This separation improves maintainability and supports incremental system growth.



Frontend Repository

Purpose: Contains the frontend source code for the web and mobile applications used by police officers, administrators, police managers, and citizens.

Structure: Organized into user interfaces, pages, services, and configuration files



Frontend–Backend Integration

Although the frontend and backend are maintained in separate repositories, both follow a consistent version control strategy, including branch management, pull requests, and release tagging. This approach enables coordinated integration while preserving the ability to develop and release each part of the system independently.

6.3 Team Members and Responsibilities

The project was developed by a team of two members. All team members collaboratively participated in the system analysis and architectural design phases,

including requirements analysis, system modeling, and defining the overall system architecture.

Backend Development:

Responsible for implementing backend services using Django REST Framework and Laravel, including user management, traffic violation management, appeal handling, database design, and integration with artificial intelligence services

Frontend Development:

Responsible for managing frontend branches, implementing user interfaces, and integrating frontend logic with backend APIs.

- AI Services Development:

responsible for implementing artificial intelligence services, including speech-to-text processing, vehicle license plate recognition and communication with backend services

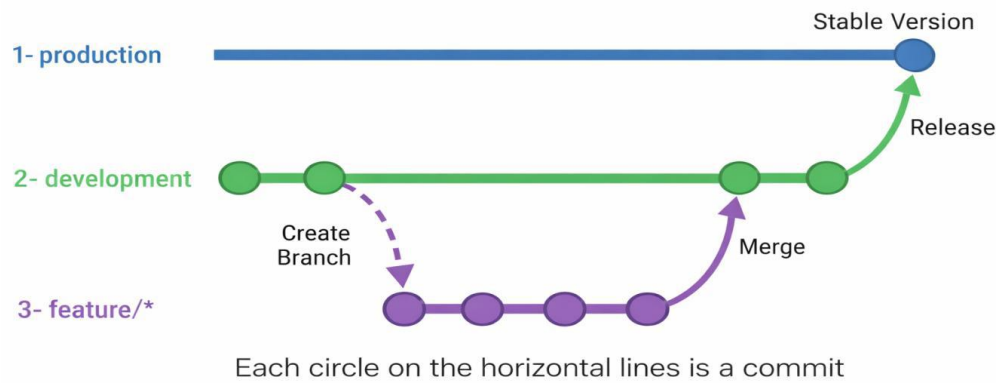
All team members contributed to version control activities. Backend- and frontend-related branches, merge operations, releases, and repository management were handled by the respective developers. Pull requests were used to merge feature branches into the development branch, ensuring controlled integration and traceability of changes.

6.4 Development Workflow

The development workflow for both the backend and frontend follows an incremental and task-based approach to ensure consistency, stability, and controlled integration across the entire system.

The workflow is summarized as follows:

1. A development task or feature is identified
2. A feature branch is created from the development branch
3. The feature is implemented and tested locally
4. A pull request is created to merge the feature branch into development
5. After integration and validation, the development branch is merged into production for release



7. Summary:

In conclusion, good project management is vital for successful software system development. It provides structure, ensuring projects are completed within scope and schedule. Overall, strong project management enhances the delivery of high quality software systems

Chapter 3

System Analysis

Introduction:

In this chapter we will introduce the analytical study of the system using the needed UML diagrams for system requirements modeling.

Requirements Elicitation:

Req-ID	Requirement Title	Description	Category	Actor	Priority
RE-FR-01	The system shall allow police officers to log in	Requires entering a valid username and password to access the system	Account Management	Police Officer	High
RE-FR-02	The system shall allow police officers to log out	Enables the police officer to safely log out of the system	Account Management	Police Officer	Medium
RE-FR-03	The system shall display and allow modification of police account information	Allows viewing and updating personal information such as phone number and email	Account Management	Police Officer	Medium
RE-FR-04	The system shall allow recording a violation by voice	Enables creating a violation using voice commands converted into text	Registering Violations	Police Officer	High
RE-FR-05	The system shall allow manual violation recording	Enables manual entry of violation details	Registering Violations	Police Officer	High
RE-FR-06	The system shall allow capturing photos of violating vehicles	Enables taking photos as evidence of violations	Registering Violations	Police Officer	High
RE-FR-07	The system shall allow review of AI-generated reports	Displays automated violation reports before approval	Report Management	Police Officer	Medium
RE-FR-08	The system shall allow modification of AI-generated reports	Allows editing reports generated by artificial intelligence	Report Management	Police Officer	Medium

RE-FR-09	The system shall display violation records created by the police officer	Shows violations with status and history	Violations Management	Police Officer	High
RE-FR-10	The system shall allow searching for violations	Enables searching violations using different criteria	Violations Management	Police Officer	Medium
RE-FR-11	The system shall allow citizens to view their violation records	Displays all violations recorded against the citizen	Violations Management	Citizen	High
RE-FR-12	The system shall allow electronic appeal submission	Enables citizens to submit and follow up appeals electronically	Appeals Management	Citizen	High
RE-FR-13	The system shall allow tracking of appeal status	Displays appeal status (under review, accepted, rejected)	Appeals Management	Citizen	High
RE-FR-14	The system shall allow police managers to log in	Enables access to the administrative control panel	Account Management	Police Manager	High
RE-FR-15	The system shall allow police managers to log out	Enables safe logout from the system	Account Management	Police Manager	Medium
RE-FR-16	The system shall allow police managers to view and edit their account information	Allows viewing and updating manager account data	Account Management	Police Manager	Medium
RE-FR-17	The system shall allow administrators to create police accounts	Enables creation of new police officer accounts	Users Management	Administrator	High
RE-FR-18	The system shall allow administrators to	Enables updating police officer account data	Users Management	Administrator	Medium

	modify police accounts				
RE-FR-19	The system shall allow administrators to delete police accounts	Enables permanent deletion of police accounts	Users Management	Administrator	Medium
RE-FR-20	The system shall allow administrators to activate or deactivate police accounts	Enables enabling or disabling police accounts	Users Management	Administrator	Medium
RE-FR-21	The system shall allow administrators to view active police accounts	Displays a list of all active police accounts	Users Management	Administrator	Medium
RE-FR-22	The system shall allow police managers to review citizen appeals	Enables reviewing appeals submitted by citizens	Appeals Management	Police Manager	Medium
RE-FR-23	The system shall convert voice commands into text	Converts spoken commands into processable text	Artificial Intelligence	AI	High
RE-FR-24	The system shall analyze captured images	Extracts vehicle plate number and type from images	Artificial Intelligence	AI	High
RE-FR-25	The system shall generate automated violation reports	Generates reports including time, location, type, and plate number	Artificial Intelligence	AI	High
RE-FR-26	The system shall classify violations	Automatically classifies violations by type	Artificial Intelligence	AI	High
RE-NFR-01	The system shall provide constant availability	Ensures continuous service availability	Availability	System	High

		with minimal downtime			
RE-NFR-02	The system shall support concurrent users	Processes multiple requests simultaneously without degradation	Performance	System	High
RE-NFR-03	The system shall be scalable	Allows the system to scale as usage increases	Scalability	System	High
RE-NFR-04	The system shall secure data and communications	Encrypts all data and communications using HTTPS	Security	System	High
RE-NFR-05	The system shall securely store passwords	Uses strong hashing algorithms	Security	System	High
RE-NFR-06	The system shall provide daily data backups	Creates automated daily database backups	Reliability	System	High

SRS Document:

Introduction:

Purpose:

The purpose of this document is to formally define the scope, architecture, and primary objectives of the Smart Traffic Police Assistant System. This system is designed to enhance the efficiency and accuracy of traffic violation management by leveraging artificial intelligence technologies, including speech-to-text processing and computer vision. The system supports traffic police officers in documenting violations through both manual and automated methods, facilitates administrative control and monitoring, and enables citizens to securely access their violation records and submit electronic appeals. Furthermore, the system aims to improve operational transparency, ensure data security, and provide a scalable and reliable digital solution that supports modern smart traffic management requirements.

Project scope:

The scope of the **Smart Traffic Police Assistant System** covers the design and implementation of a digital platform that supports traffic violation management through integrated artificial intelligence services. The system focuses on enabling traffic police officers to record violations using voice commands, manual data entry, and image capture, while automatically generating and classifying violation reports. It also includes administrative functionalities for managing police accounts, reviewing citizen appeals, and monitoring system operations. Additionally, the system provides citizens with secure access to their violation records and electronic appeal services. The scope of this project is limited to the software-based management of traffic violations and does not include physical traffic enforcement hardware or real-time traffic signal control.

Document overview:

The software requirements specification SRS document will be structured as follows:

1. Introduction.
2. Overall description and product features.
3. Nonfunctional requirements.
4. Conclusion.

Overall description and product features:

Product features:

The product features derived from the system's use cases include:

User Authentication (UC-01):

Allows police officers, police managers, administrators, and citizens to log in and log out of the system securely using authenticated credentials.

Manage Police Profile (UC-02):

Enables police officers to view and edit their profile information, such as personal and contact details.

Manage Police Accounts (UC-03):

Allows administrators to create, edit, activate, or deactivate police officer accounts within the system.

Record Violation Manually (UC-04):

Enables police officers to manually add a new traffic violation by entering violation details through the system.

Record and Transcribe Voice Commands (UC-05):

Allows police officers to record violation details using voice commands, which are automatically converted into text.

Capture and Analyze Vehicle Images (UC-06):

Enables capturing images of violating vehicles and extracting license plate information using image processing techniques.

Generate Violation Report (UC-07):

Automatically generates a structured violation report by combining manual input, voice transcription, and image analysis.

Manage Violations (UC-08):

Allows police officers to view, manage, and track recorded violations and their statuses.

Review Violations (UC-09):

Enables police managers to review recorded violations for monitoring and administrative purposes.

Submit Appeal (UC-10):

Allows citizens to submit an electronic appeal against a traffic violation.

Review Appeals (UC-11):

Enables police managers to review, accept, or reject appeals submitted by citizens.

View Violations for Citizens (UC-12):

Allows citizens to view their personal traffic violation records through the system

user class and charactrtistics

The system supports the following user classes:

Police Officer:

Primary users responsible for recording traffic violations, capturing evidence, generating violation reports, and managing violation records. They require moderate technical skills and use the system mainly in field operations.

Citizen:

End users who access the system to view their traffic violations and submit appeals. They require minimal technical knowledge.

Police Manager:

Supervisory users responsible for reviewing violations and handling citizen appeals. They require administrative and decision-making capabilities.

System Administrator:

Privileged users responsible for managing police accounts, including creating, editing, activating, and deactivating accounts. This role does not interact directly with violation data.

Assumption

Internet Connectivity:

It is assumed that users will have access to a stable internet connection, as the system operates as an online platform.

Mobile Usage:

The system is designed primarily for mobile use, especially for police officers working in the field.

Authorized System Access:

All users are assumed to access the system using valid credentials assigned according to their roles.

Accuracy of Data Entry:

Police officers are expected to enter accurate and complete violation information to ensure reliable reports and system outputs.

AI Assistance Limitation:

Outputs generated by artificial intelligence components are considered supportive and subject to human review before final approval.

Non functional requirements:

The non-functional requirements for the Smart Traffic Police Assistant System are as follows:

Usability:

The system shall provide user-friendly interfaces that enable traffic police officers, administrators, and citizens to perform core tasks efficiently. This includes simple and intuitive workflows for recording violations, reviewing reports, and submitting appeals, as well as clear visual cues such as meaningful icons and guided input forms to reduce user errors during field operations.

Security:

The system shall protect sensitive user data, including account information, violation records, and appeal details, by implementing secure authentication mechanisms, role-based access control, and encrypted communication channels (HTTPS). Additionally, an API Gateway shall be utilized to enforce access control policies and isolate backend services and databases from unauthorized access.

Conclusion:

This software Requirements Specification (SRS) document outlines the critical functionalities, user roles, and assumptions essential for developing a robust and user-friendly system. By adhering to these requirements, the system will effectively support software product owners in enhancing their products to meet user needs and streamline project management processes.

Requirements modeling:

Use case diagram :

use-case diagrams model the behavior of a system and help to capture the requirements of the system.



Figure 1 Use case diagram

Uc -1-

use case specification:

Use case number	Uc_1
Title	Login / Logout
Description	This feature allows police officers and administrators to securely log in and out of the system.
Actors	Police Officer, Administrator
Precondition	The user must have an active account.
Main Flow	1. The officer selects 'Login'. 2. The system displays the login form. 3. The user enters the username and password. 4. The system verifies the credentials. 5. If valid, access is granted. 6. To log out, the user selects 'Logout' and the session ends.
Alternative flows	First alternative flow A1: Start at step 4 in the main flow where the credentials are invalid. 5. The system displays an error message and returns to step 2.
Postcondition	The user session is successfully created or terminated.

sequence diagram

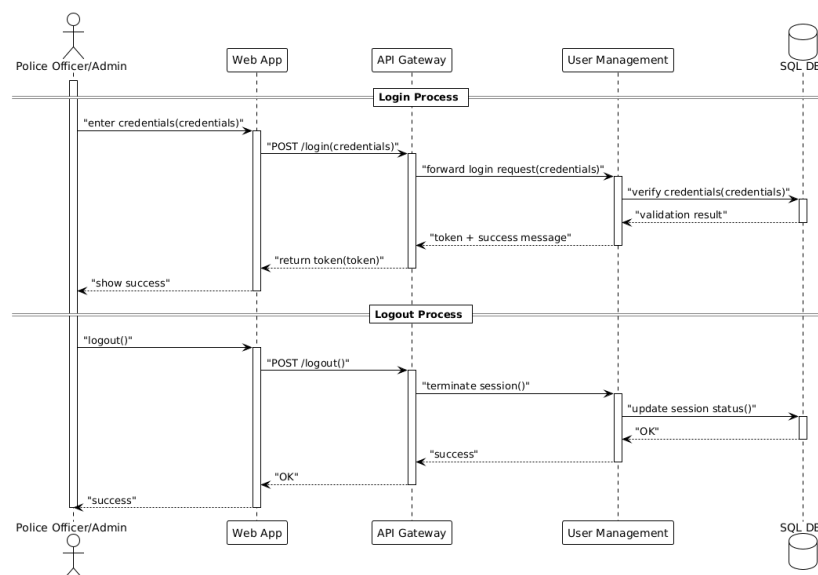


Figure 2 sequence diagram “Login / Logout”

Uc -2-

use case specification:

Use case number	Uc_2
Title	Account Management
Description	This feature allows the police officer to manage and update personal information.
Actors	Police Officer
Precondition	The officer must be logged in.
Main Flow	1. The officer selects 'View Profile'. 2. The system displays the account details. 3. if the officer need selects 'Edit Information'. 4. The system displays the edit form. 5. The officer enters the new data. 6. The system validates and saves the updates. 7. The system displays a success message.
Alternative flows	First alternative flow A1: Start at step 6 where invalid data is entered. 7. The system displays an error message and returns to step 4.
Postcondition	The officer's profile information is updated successfully or the user is informed of any failure

Activity diagram:

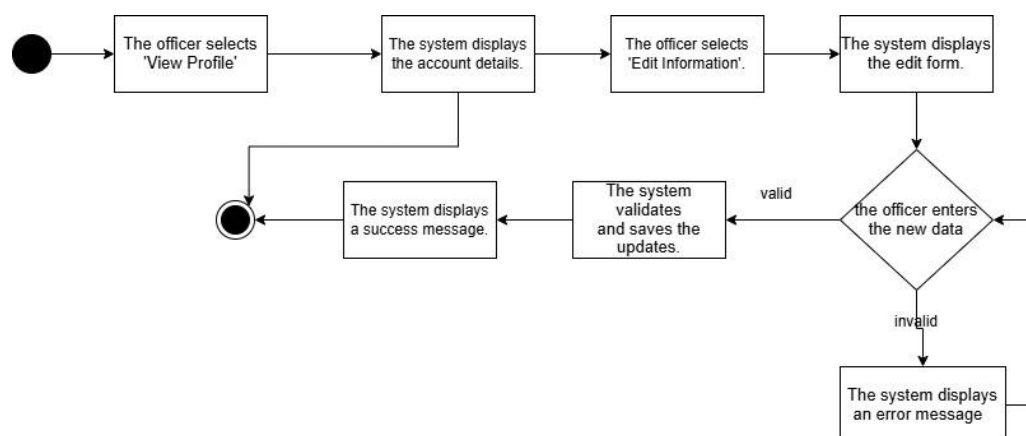


Figure 3 activity diagram “Account Management”

sequence diagram

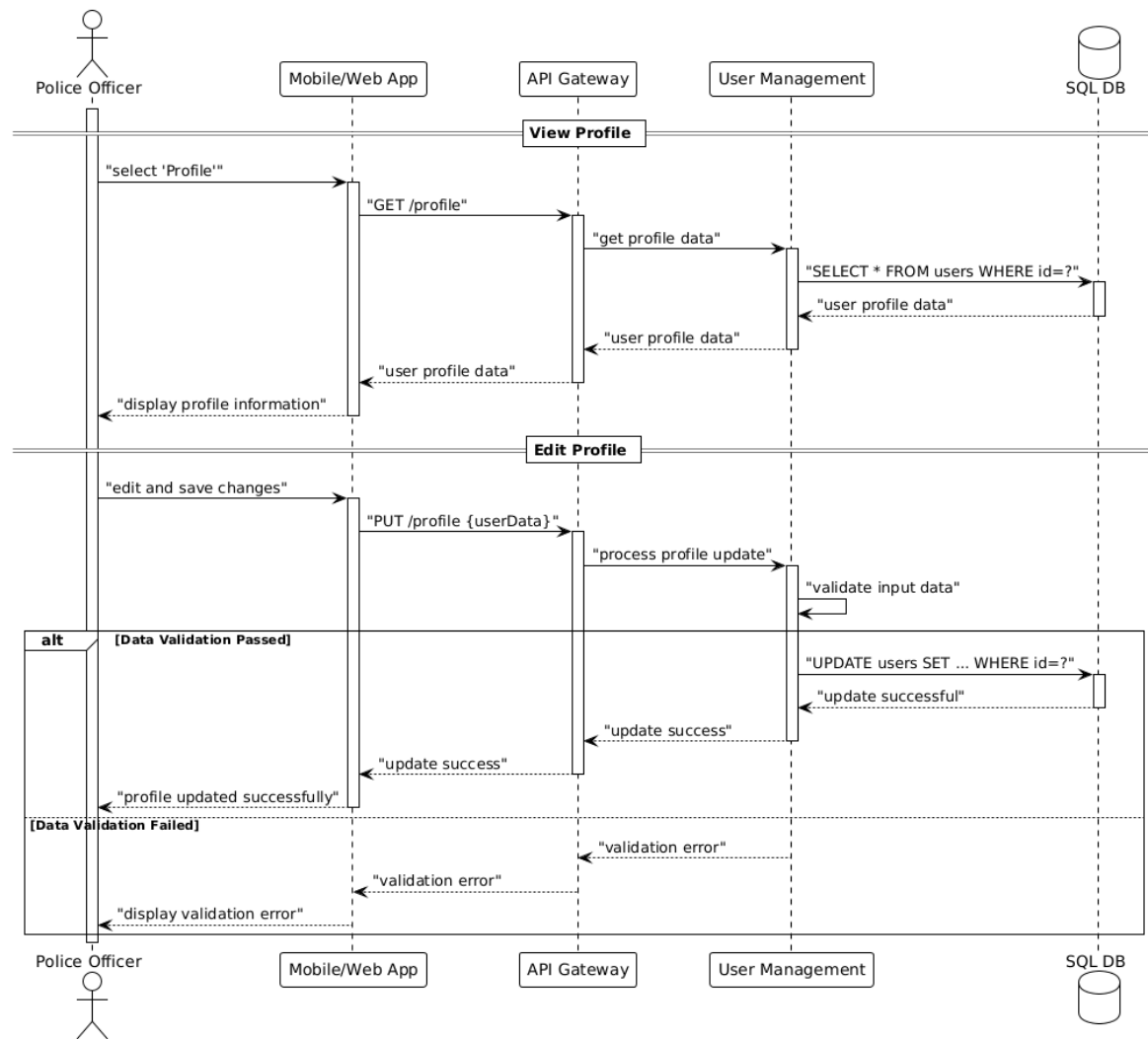


Figure 4 sequence diagram "Account Management"

Uc -3-

use case specification:

use case number	UC_3
title	Generate Violation
description	This feature allows the police officer to create an AI-assisted traffic violation record.
actors	Police Officer
precondition	The officer must be logged in.
main flow	The officer selects 'Generate Violation'. 2. The system prompts to record audio. 3. The officer records the audio. 4. The system converts it to text and displays it. 5. The officer captures an image of the vehicle. 6. The system extracts the license plate number. 7. The officer selects the violation type. 8. The system generates the violation. 9.the officer review the violation 10. the officer can edit the generated violation 11. The officer confirms 12. the system save the generated violation.
postcondition	The violation is successfully created or temporarily saved for later submission.
alternative flow	First alternative flow A1: Start at step 3 where the recording fails. 4. The system requests re-recording. Second alternative flow A2: Start at step 4 where conversion fails. 5. The system allows manual text entry. Third alternative flow A3: Start at step 6 where extraction fails. 7. The system asks to enter the plate number manually.

Activity diagram:

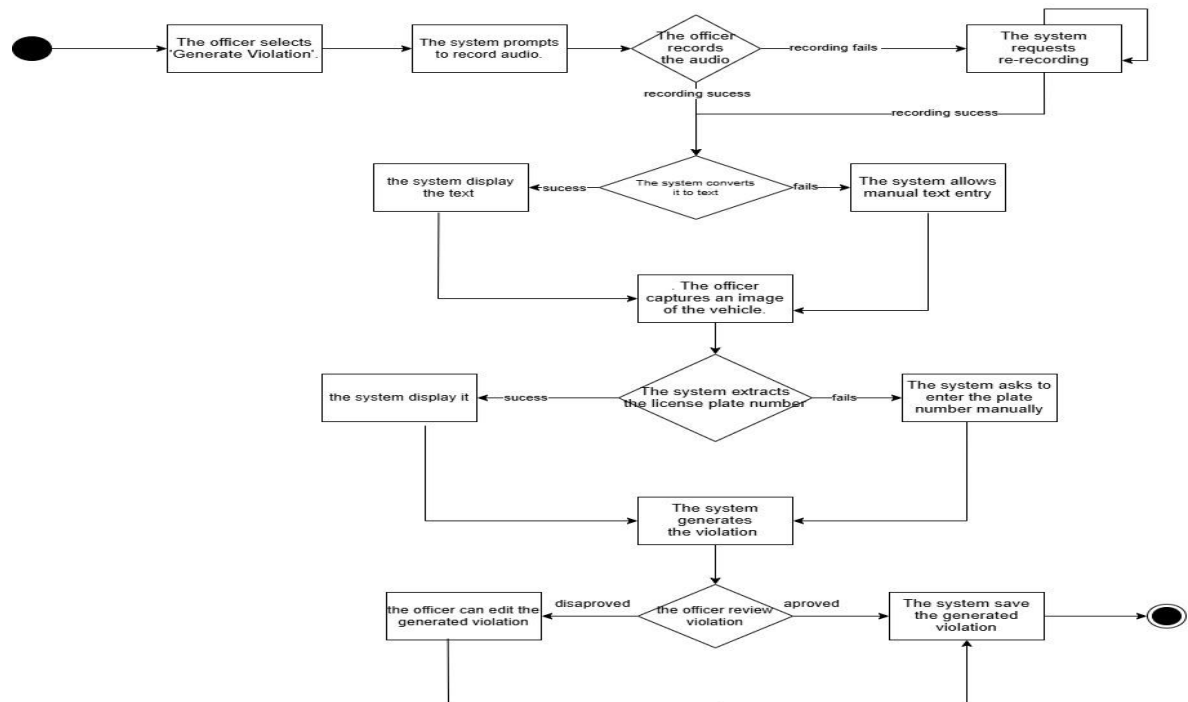


Figure 5 activity diagram “Generate Violation”

sequence diagram

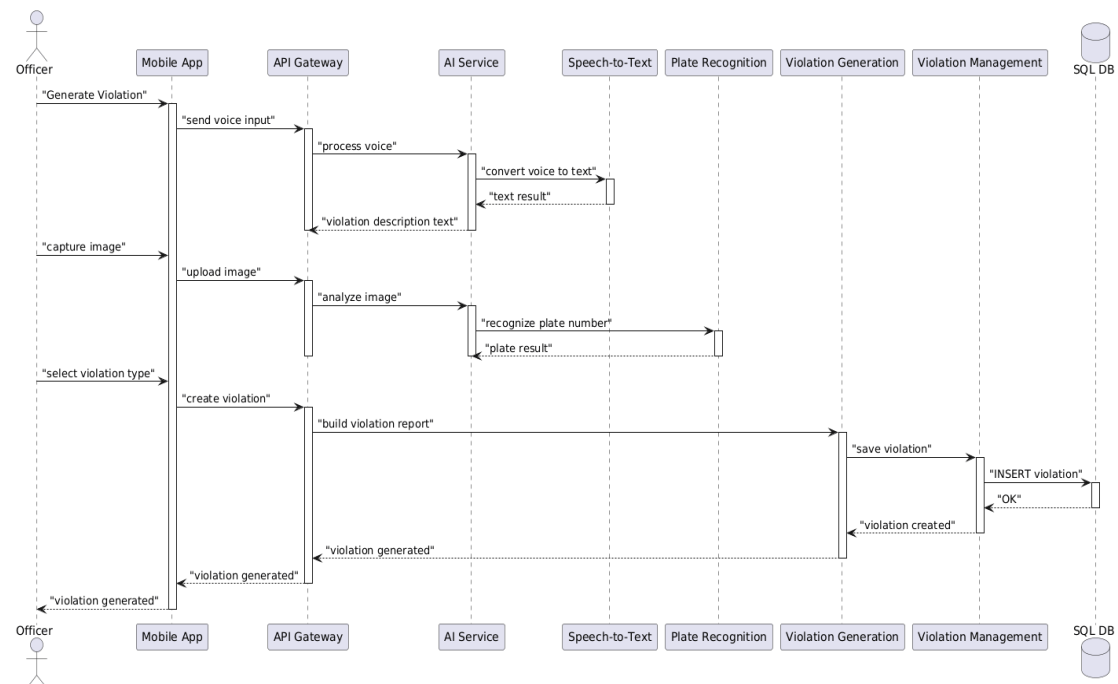


Figure 6 sequence diagram “Generate Violation”

Uc -4-

use case specification:

Use case number	UC_4
title	Add Violation
description	This feature allows the police officer to manually add a new traffic violation.
actors	Police Officer
precondition	The officer must be logged in.
main flow	1. The officer selects 'Add Violation'. 2. The system displays a data entry form. 3. The officer enters violation details. 4. The system validates the data. 5. The system generates and displays the violation report. 6. The officer reviews and confirms.
postcondition	The violation is successfully saved.
alternative flow	First alternative flow A1: Start at step 4 where data is invalid or missing. 5. The system displays an error and returns to step 3.

Activity diagram:

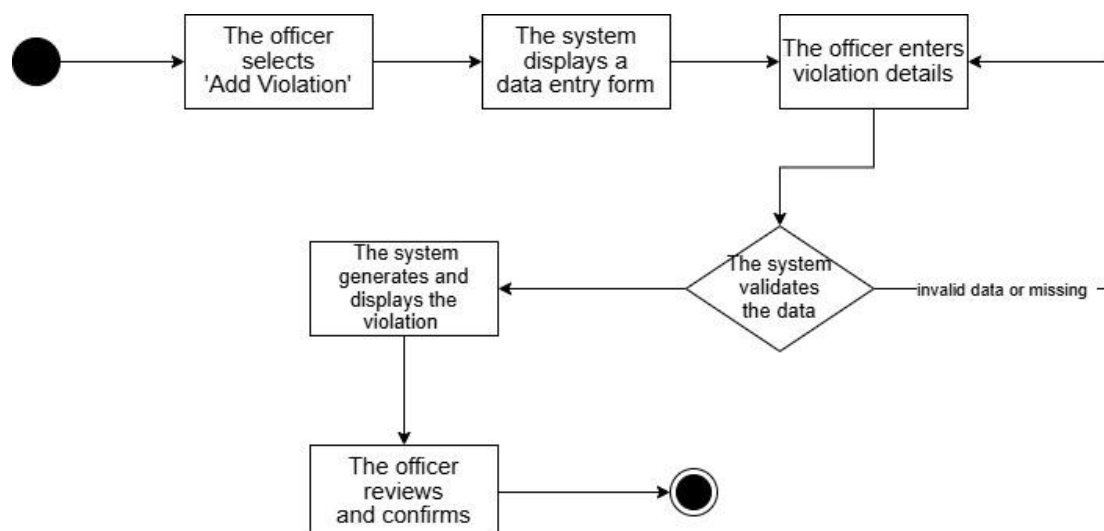


Figure 7 activity diagram “Add Violation”

Uc -5-

use case specification:

Use case number	UC_5
title	View Violation
description	This feature allows the police officer to view and manage existing violations.
actors	Police Officer
precondition	The officer must be logged in and have previous violations.
main flow	1. The officer selects 'View Violations'. 2. The system displays the list of violations. 3. If the officer need the can search by date or plate. 4. The system displays details.
alternative flow	First alternative flow A1: Start at step 2 where no violations exist. 3. The system displays 'No violations found'.
postcondition	The violation details are successfully displayed.

Activity diagram:

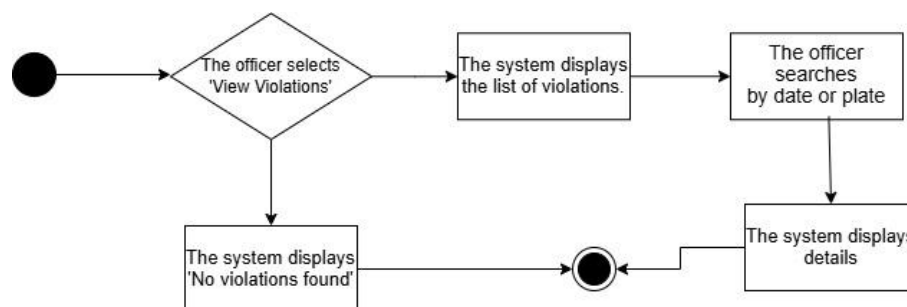


Figure 8 activity diagram “View Violation”

sequence diagram:

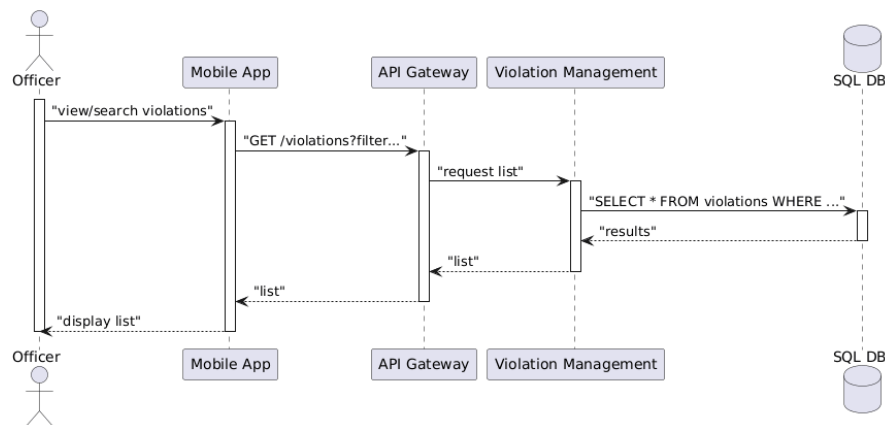


Figure 9 sequence diagram “View Violation

Uc -6-

use case specification:

Use case number	UC_6
title	Create Police Account
description	This feature allows the administrator to create new police officer accounts.
actors	Administrator
precondition	The administrator must be logged in.
main flow	1. The administrator selects 'Create New Account'. 2. The system displays the registration form. 3. The administrator enters officer details. 4. The system validates and saves the data. 5. The system confirms creation.
postcondition	The new police account is successfully created.
Alternative flow	First alternative flow A1: Start at step 4 where data is incomplete or duplicate. 5. The system displays an error message and returns to step 3.

Activity diagram:

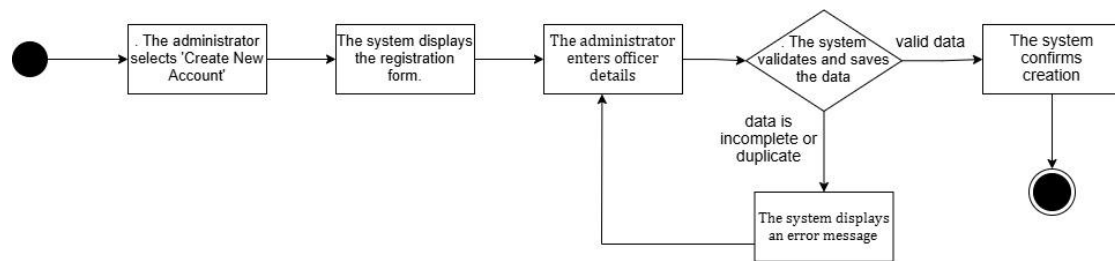


Figure 10 activity diagram “Create Police Account”

sequence diagram:

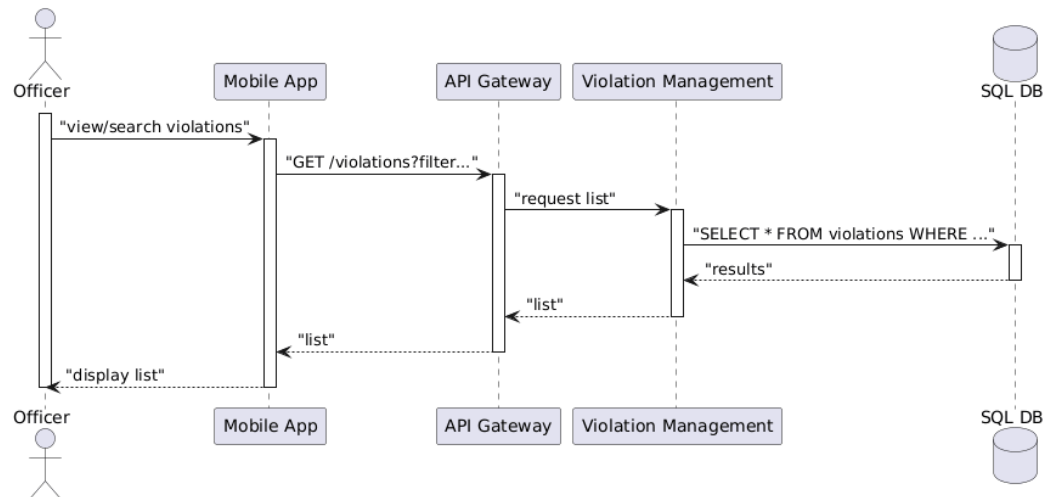


Figure 11 sequence diagram “Create Police Account”

Uc -7-

use case specification:

Use case number	uc_7
title	Edit Police Account
description	This feature allows the administrator to edit existing police officer account information.
actors	Administrator
precondition	The administrator must be logged in.
main flow	1. The administrator selects 'Manage Police Accounts'. 2. The system displays the list. 3. The administrator selects an account. 4. The system displays account details. 5. The administrator edits and saves changes.
postcondition	The police account information is updated successfully.
Alternative flow	First alternative flow A1: Start at step 6 where validation fails. 7. The system displays an error and requests correction.

Uc -8-

use case specification:

Use case number	Uc_8
Title	Activate / Deactivate Police Account
Description	This feature allows the administrator to enable or disable police officer accounts.
Actors	Administrator
Precondition	The administrator must be logged in.
Main Flow	1. The administrator selects 'Manage Police Accounts'. 2. The system displays accounts. 3. The administrator selects an account. 4. The administrator confirms activation or deactivation. 5. The system updates the status.
Postcondition	The selected police account is activated or deactivated successfully.

Activity diagram:

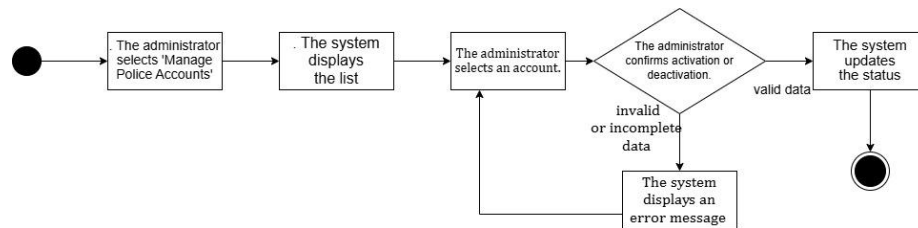


Figure 12 activity diagram “Activate / Deactivate Police Account”

Uc -9-

use case specification:

Use case number	Uc_9
Title	Review violation
Description	This feature allows the Police manager to review and audit violations submitted by police officers.
Actors	Police manager
Precondition	The Police manager must be logged in.
Main Flow	1. The police manager selects 'Review Violations'. 2. The system displays the list. 3. The police manager searches and views a specific violation.
Postcondition	The police manager has successfully reviewed the violation.
Alternative flow	First alternative flow A1: Start at step 3 where no matching violation is found. 4. The system displays a message 'No matching violations found'.

sequence diagram:

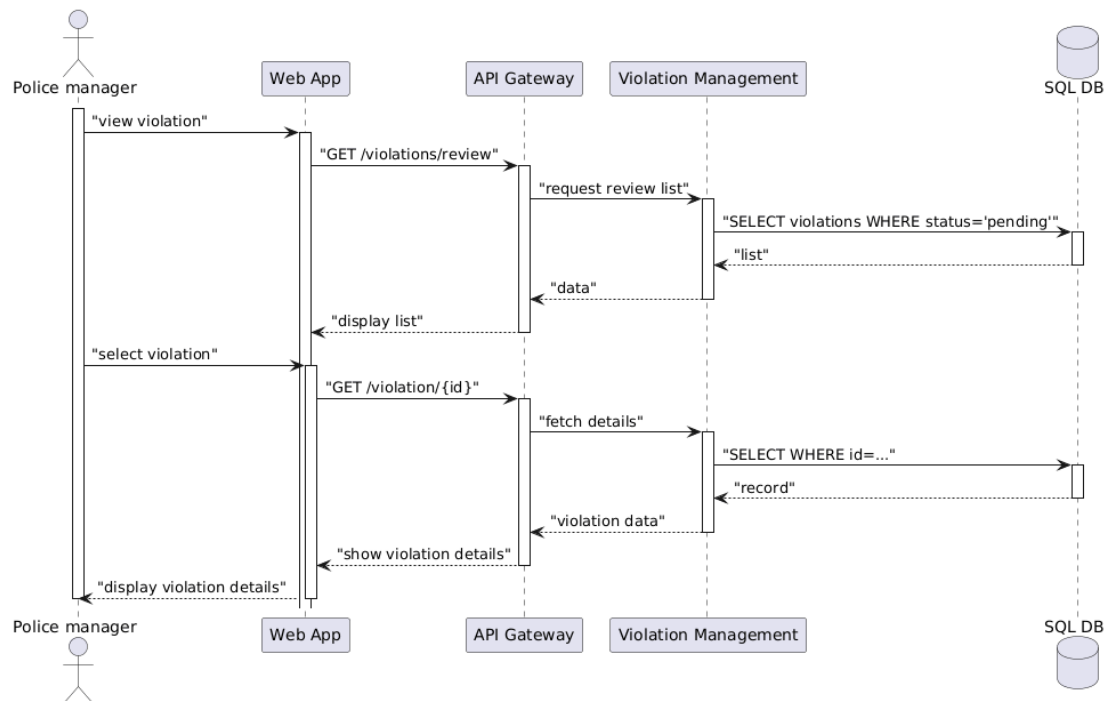


Figure 13 sequence diagram “Review violation”

Uc -10-

use case specification:

Use case number	Uc_10
Title	Review appeals
Description	This feature allows the administrator to review and make decisions on citizen appeals.
Actors	Police manager
Precondition	The police manager must be logged in.
Main Flow	1. The police manager selects 'Review appeals.' 2. The system displays the list. 3. The police manager opens an appeal. 4. The system displays its details. 5. The police manager decides to accept or reject. 6. The system records the decision.
Alternative flows	First alternative flow A1: Start at step 6 where decision submission fails. 7. The system displays 'Error submitting decision'.
Postcondition	The appeal status is updated successfully.

sequence diagram:

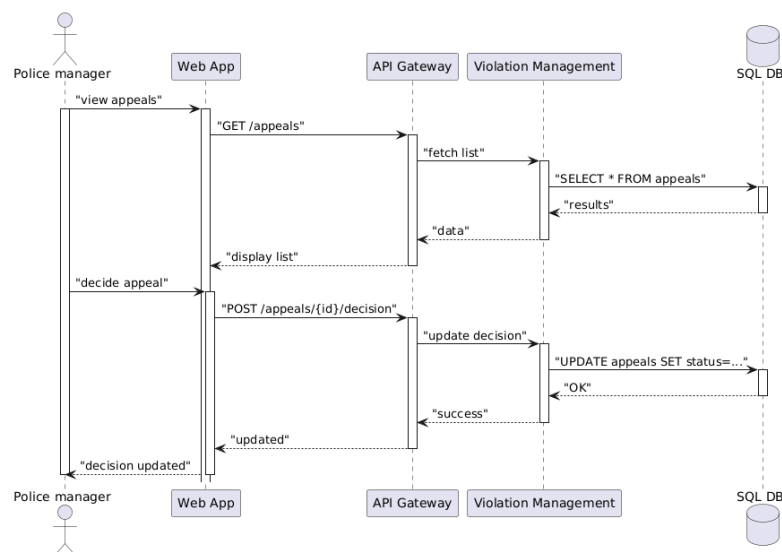


Figure 14 sequence diagram “Review appeals”

Uc -11-

use case specification:

Use case number	Uc_11
Title	View Violations
Description	This feature allows citizens to view their traffic violations.
Actors	Citizen
Precondition	The citizen enters a valid vehicle number.
Main Flow	1. The citizen enters vehicle number. 2. The system verifies the data. 3. The system displays the violations list. 4. The citizen selects a violation. 5. The system displays its details.
Alternative flows	First alternative flow A1: Start at step 2 where invalid data is entered. 3. The system displays an error message. Second alternative flow A2: Start at step 3 where no violations exist. 4. The system displays 'No violations found'.
Postcondition	The citizen successfully views all related violations.

Uc -12-

use case specification:

Use case number	Uc_12
Title	Submit appeal
Description	This feature allows the citizen to submit an appeal on a specific violation.
Actors	Citizen
Precondition	The citizen must have viewed the violation beforehand.
Main Flow	1. The citizen selects 'Submit appeal. 2. The system displays the appeal form. 3. The citizen enters the reason and notes. 4. The system validates and saves the appeal.
Alternative flows	First alternative flow A1: Start at step 4 where data is incomplete. 5. The system displays 'Incomplete data'. Second alternative flow A2: Start at step 4 where a technical error occurs. 6. The system displays 'Unable to submit appeal currently'.
Postcondition	The appeal is successfully submitted.

sequence diagram:

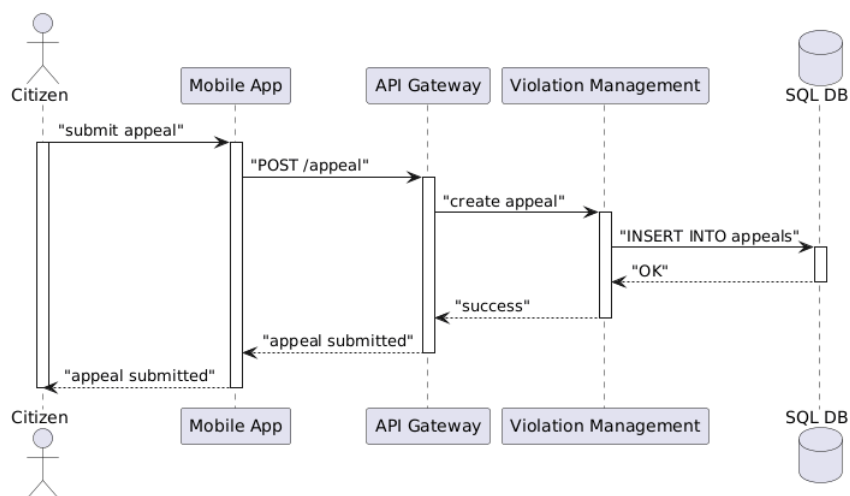


Figure 15 sequence diagram “Submit appeal”

Class case diagram :

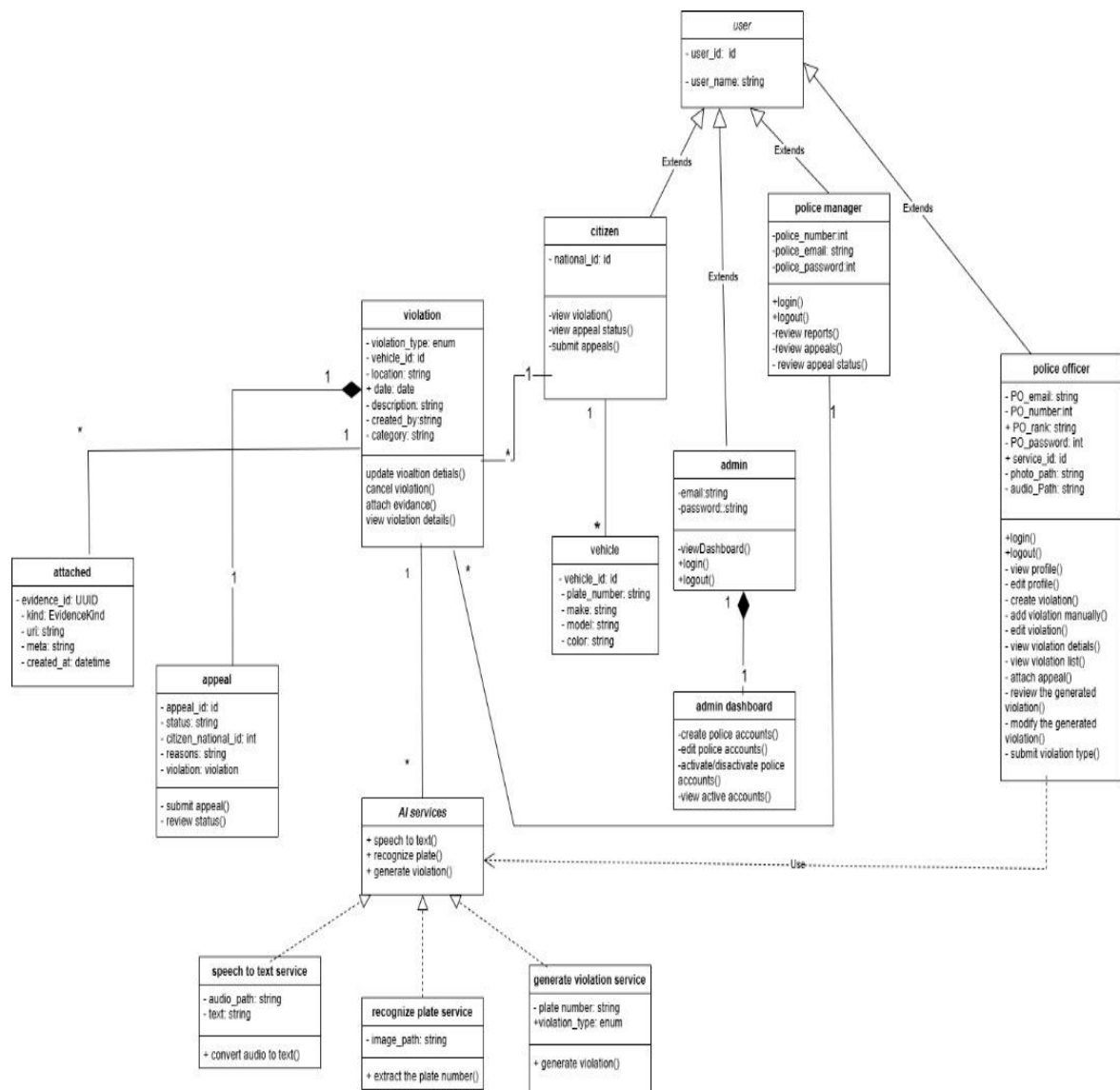


Figure 16 class diagram

ERD – Entity Relationship Diagram :

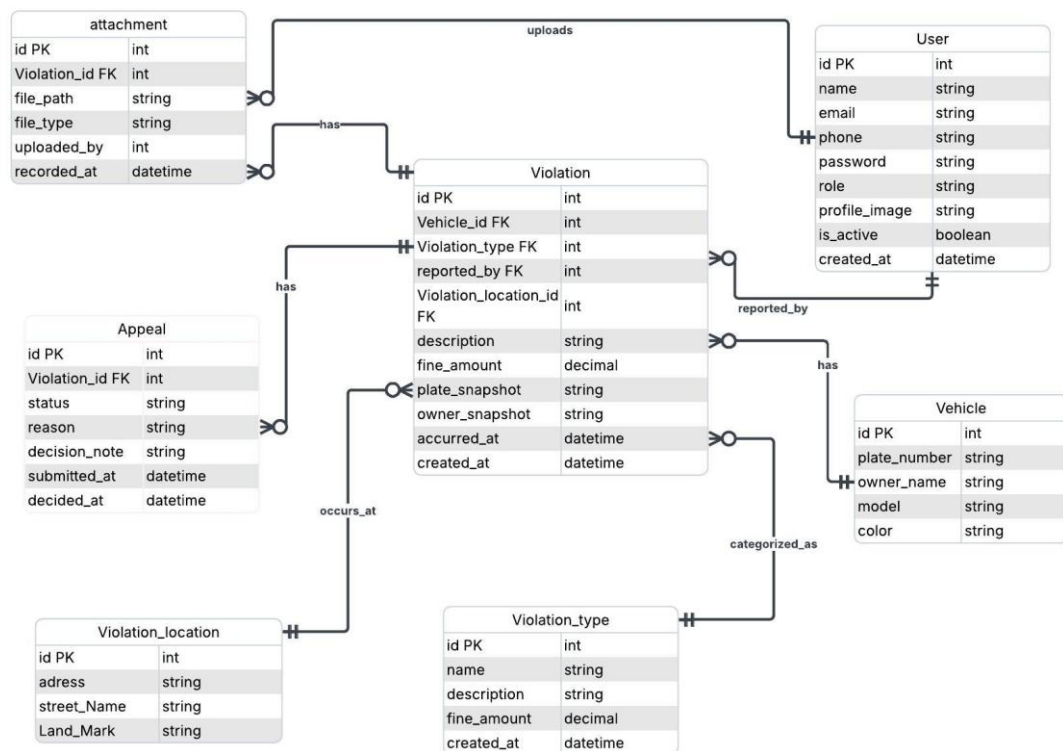


Figure 17 ERD

Initial Test Cases:

Test Case Scenario: Sce-01: login/logout.

Test case id	Test case title	Test steps	Expected result
UC1-TC01	Check successful login with valid credentials	1. Launch the application on the login page.2. Enter a valid username and password.3. Select Login.	Login is successful. The system grants access to the system main page. A user session is created.
UC1-TC02	Check login fails with missing fields	1. Launch the application on the login page.2. Leave the username and/or password field empty.3. Select Login.	The system displays an error message indicating a missing field and remains on the login page.
UC1-TC03	Check login fails with invalid credentials (non-existent username)	1. Launch the application on the login page.2. Enter a non-existent username and any password.3. Select Login.	The system displays an error message such as "Invalid username or password" and returns to the login form.
UC1-TC04	Check login fails with invalid credentials (incorrect password)	1. Launch the application on the login page.2. Enter a valid username and an incorrect password.3. Select Login.	The system displays an error message such as "Invalid username or password" and returns to the login form.
UC1-TC05	Check successful logout	1. Log in with valid credentials.2. Select Logout.	The user is logged out successfully. The session ends and the system returns to the login page.

Test Case Scenario: Sce-02: Account Management:

Test case id	Test case title	Test steps	Expected result
UC2-TC01	Check view account profile	1. Log in as a police officer.2. Select View Profile.	The system displays the officer's account details correctly.
UC2-TC02	Check successful edit of account information with valid data	1. Log in as a police officer.2. Select View Profile.3. Select Edit Information.4. Enter valid new data.5. Select Save.	The system validates and saves the data successfully and displays a success message.

UC2-TC03	Check edit account information with invalid data	1. Log in as a police officer.2. Select View Profile.3. Select Edit Information.4. Enter invalid or incomplete data.5. Select Save.	The system displays an error message and returns to the edit information form.
UC2-TC04	Check edit account information without making changes	1. Log in as a police officer.2. Select View Profile.3. Select Edit Information.4. Do not modify any fields.5. Select Save.	The system keeps the existing data unchanged and displays an appropriate confirmation or no-change message.

Test Case Scenario: Sce-03: Generate violation:

Test case id	Test case title	Test steps	Expected result
UC4-TC01	Check successful income recording with valid data.	1. Navigate to the "Manage Incomes" page. 2. Choose to "record incomes". 3. Fill in a valid amount. 4. Select a valid date. 5. Choose an income category.	The system displays "the incomes has successfully recorded".
UC4-TC02	Check income recording fails with invalid amount.	1. Navigate to the "Manage Incomes" page. 2. Choose to "record incomes". 3. Enter an invalid amount (e.g., negative, zero, empty). 4. Select a valid date. 5. Choose an income category.	The system will show an error message: "Please enter a valid amount."
UC4-TC03	Check income recording fails with empty date field.	1. Navigate to the "Manage Incomes" page. 2. Choose to "record incomes". 3. Fill in a valid amount. 4. Leave the date field empty.	The system will display a message: "Please select a date for the income."

		5. Choose an income category.	
UC4-TC04	Check income recording fails with no category selected.	1. Navigate to the "Manage Incomes" page. 2. Choose to "record incomes". 3. Fill in a valid amount. 4. Select a valid date. 5. Do not choose an income category.	The system will show a message: "Please choose a category for the income."

Test Case Scenario: Sce-04: Add Violation.

Test case id	Test case title	Test steps	Expected result
UC4-TC01	Check successful addition of a violation with valid data	1. Log in as a police officer. 2. Select Add Violation . 3. Enter valid violation details. 4. Confirm the violation.	The system validates the data, generates the violation report, and saves the violation successfully.
UC4-TC02	Check add violation with missing required fields	1. Log in as a police officer. 2. Select Add Violation . 3. Leave one or more required fields empty. 4. Submit the form.	The system displays an error message and returns to the violation data entry form.
UC4-TC03	Check add violation with invalid data	1. Log in as a police officer. 2. Select Add Violation . 3. Enter invalid data (e.g., incorrect format). 4. Submit the form.	The system displays an error message and allows the officer to correct the entered data.

Test Case Scenario: Sce-05: view violation.

Test case id	Test case title	Test steps	Expected result
UC5-TC01	Check viewing list of existing violations	1. Log in as a police officer with existing violations. 2. Select View Violations .	The system displays the list of existing violations successfully.
UC5-TC02	Check viewing violation details	1. Log in as a police officer with existing violations. 2. Select View Violations . 3. Select view violation details.	The system displays the detailed information of the selected violation.
UC5-TC03	Check searching violations by date	1. Log in as a police officer with existing violations. 2. Select View Violations . 3. Search for violations using a date. 4. Select a violation from the results.	The system displays the violations that match the selected date and shows the violation details.
UC5-TC04	Check searching violations by license plate number	1. Log in as a police officer with existing violations. 2. Select View Violations . 3. Search for violations using a license plate number. 4. Select a violation from the results.	The system displays the violations that match the entered plate number and shows the violation details.
UC5-TC05	Check viewing violations when none exist	1. Log in as a police officer with no previous violations. 2. Select View Violations .	The system displays the message "No violations found".

Test Case Scenario: Sce-06: Create Police Account

Test case id	Test case title	Test steps	Expected result
UC6-TC01	Check successful creation of a police account with valid data	1. Log in as an administrator. 2. Select Create New Account . 3. Enter valid and complete police officer details. 4. Submit the registration form.	The system validates the data, saves the new police account, and displays a successful creation confirmation.
UC6-TC02	Check police account creation with	1. Log in as an administrator. 2. Select Create New Account . 3. Leave one or more required	The system displays an error message and returns to the

UC6-TC03	missing required data Check police account creation with duplicate data	fields empty.4. Submit the registration form. 1. Log in as an administrator.2. Select Create New Account .3. Enter officer details that already exist (e.g., duplicate username or officer ID).4. Submit the registration form.	registration form to correct the data. The system displays an error message indicating duplicate data and allows the administrator to re-enter valid information.
UC6-TC04	Check police account creation with invalid data	1. Log in as an administrator. 2. Select Create New Account . 3. Enter invalid data (e.g., incorrect email format, invalid ID format). 4. Submit the registration form	The system displays an error message indicating invalid data and returns to the registration form for correction

Test Case Scenario: Sce-07: edit police account.

Test case id	Test case title	Test steps	Expected result
UC7-TC01	Check successful editing of police account with valid data.	1. Log in as Administrator. 2. Select Manage Police Accounts . 3. Choose an existing police account from the list. 4. Edit account details with valid data. 5. Save the changes	The system validates the data and updates the police account successfully. A confirmation message is displayed
UC7-TC02	Check editing police account with missing required fields	1. Log in as Administrator. 2. Select Manage Police Accounts . 3. Choose an existing police account. 4. Remove or leave one or more required fields empty. 5.	The system displays an error message indicating missing required fields and requests correction
UC7-TC03	Check editing police account with invalid data format	1. Log in as Administrator. 2. Select Manage Police Accounts . 3. Choose an existing police account. 4. Enter invalid data (e.g., incorrect phone number format). 5. Save the changes	The system rejects the input and displays a validation error message.

Test Case Scenario: Sce-08: Activate / Deactivate Police Account :

Test case id	Test case title	Test steps	Expected result
UC8-TC01	Check successful activation of a police account	1. Log in as Administrator. 2. Select Manage Police Accounts. 3. Choose a deactivated police account. 4. Confirm Activate action	The system activates the police account successfully and updates the account status to <i>Active</i>
UC8-TC02	Check successful deactivation of a police account	1. Log in as Administrator. 2. Select Manage Police Accounts. 3. Choose an active police account. 4. Confirm Deactivate action	The system deactivates the police account successfully and updates the account status to <i>Inactive</i>
UC8-TC03	Check activation/deactivation cancellation	1. Log in as Administrator. 2. Select Manage Police Accounts. 3. Choose any police account. 4. Click Activate/Deactivate. 5. Cancel the confirmation dialog	The system cancels the operation and the police account status remains unchanged
UC8-TC04	Check status update reflected in police accounts list	1. Log in as Administrator. 2. Select Manage Police Accounts. 3. Activate or deactivate a police account. 4. Return to the accounts list	The updated account status is correctly reflected in the police accounts list

Test Case Scenario: Sce-09: Review Violation:

Test case id	Test case title	Test steps	Expected result
UC9-TC01	Check display of current month's total expenses and income.	1. Log in as Police Manager. 2. Select Review Violations. 3. The system displays the list of violations. 4. Search for an existing violation.	The system displays the violation details successfully and the violation is reviewed

		5. Select and view the violation details	
UC9-TC02	Check search for a specific violation with valid criteria	1. Log in as Police Manager. 2. Select Review Violations . 3. Enter valid search criteria (e.g., violation ID). 4. Submit search	The system displays the matching violation in the list
UC9-TC03	Check review violation details	1. Log in as Police Manager. 2. Select Review Violations . 3. Choose a violation from the list	The system displays full details of the selected violation.
UC9-TC04	Check system behavior when no matching violation is found	1. Log in as Police Manager. 2. Select Review Violations . 3. Enter invalid or non-existing search criteria. 4. Submit search.	The system displays a message " No matching violations found "

Test Case Scenario: Sce-10: Review Appeals:

Test case id	Test case title	Test steps	Expected result
UC10-TC01	Check successful review and acceptance of an appeal	1. Log in as Police Manager. 2. Select Review Appeals . 3. The system displays the appeals list. 4. Open an appeal. 5. Review appeal details. 6. Choose Accept . 7. Submit decision	The system records the decision successfully and updates the appeal status to <i>Accepted</i>
UC10-TC02	Check successful review and rejection of an appeal	1. Log in as Police Manager. 2. Select Review Appeals . 3. The system displays the appeals list. 4. Open an appeal. 5. Review appeal details. 6. Choose Reject . 7. Submit decision	The system records the decision successfully and updates the appeal status to <i>Rejected</i>

UC10-TC03	Check display of appeals list	1. Log in as Police Manager. 2. Select Review Appeals	The system displays full details of the selected appeal
UC10-TC04	Check display of appeal details	1. Log in as Police Manager. 2. Select Review Appeals . 3. Open an appeal from the list	The system sends a message "try again" or similar error message.
UC10-TC05	Check system behavior when decision submission fails	1. Log in as Police Manager . 2. Select Review Appeals . 3. Open an appeal from the appeals list. 4. Choose Accept or Reject . 5. Submit the decision. 6. A system failure occurs (e.g., database error or server error).	The system will show a message "no active accounts available".
UC10-TC06	Check decision confirmation is saved	1. Log in as Police Manager. 2. Review an appeal. 3. Accept or reject the appeal. 4. Return to appeals list	The system displays the error message: "Error submitting decision"

Test Case Scenario: Sce-11: view violation :

Test case id	Test case title	Test steps	Expected result
UC11-TC01	Check successful viewing of violations with valid vehicle number	1. Enter a valid vehicle number. 2. The system verifies the entered data. 3. The system displays the list of violations. 4. Select a violation from the list. 5. View violation details	The system displays the violations list and shows the selected violation details successfully
UC11-TC02	Check system behaviour with invalid vehicle number	1. Enter an invalid or incorrectly formatted vehicle number. 2. Submit the data.	The system displays an error message indicating invalid vehicle number and no violations are shown
UC11-TC03	Check system behavior when no violations exist	1. Enter a valid vehicle number with no associated violations. 2. Submit the data	The system displays the message "No violations found"
UC11-TC04	Check display of violations list	1. Enter a valid vehicle number. 2. Submit the data	The system displays a list of all violations related to the entered vehicle number

UC11-TC05	Check display of violation details	1. Enter a valid vehicle number. 2. View the violations list. 3. Select a violation.	The system displays full details of the selected violation
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Test Case Scenario: Sce-12: submit Appeals:

Test case id	Test case title	Test steps	Expected result
UC12-TC01	Check successful submission of an appeal with valid data	1. View a specific violation. 2. Select Submit Appeal. 3. The system displays the appeal form. 4. Enter valid reason and notes. 5. Submit the appeal	The system validates the data, saves the appeal successfully, and displays a confirmation message
UC12-TC02	Check submission of appeal with incomplete data	1. View a specific violation. 2. Select Submit Appeal. 3. The appeal form is displayed. 4. Leave required fields empty. 5. Submit the appeal.	The system displays the message "Incomplete data" and the appeal is not saved
UC12-TC03	Check system behavior when a technical error occurs during appeal submission	1. View a specific violation. 2. Select Submit Appeal. 3. Enter valid appeal data. 4. Submit the appeal. 5. A technical/system error occurs	The system displays the message "Unable to submit appeal currently" and the appeal is not saved
UC12-TC04	Check display of appeal submission form	1. View a specific violation. 2. Select Submit Appeal	The system displays the appeal submission form correctly
UC12-TC05	Check that appeal is linked to the correct violation	1. View a specific violation. 2. Submit an appeal for that violation	The system links the submitted appeal to the selected violation correctly

Initial RTM Requirements Trackability Matrix:

Req-ID	Use Case	Analysis	System Design	Detailed Design	Coding	Test Case(s)
RE-FR-01	UC-1 (Login / Logout)	AM-AN				UC1-TC01, UC1-TC02
RE-FR-02	UC-1 (Login / Logout)	AM-AN				UC1-TC03
RE-FR-03	UC-2 (Account Management)	AM-AN				UC2-TC01, UC2-TC02
RE-FR-04	UC-3 (Generate Violation)	AM-AN				UC3-TC01
RE-FR-05	UC-4 (Add Violation)	AM-AN				UC4-TC01, UC4-TC02
RE-FR-06	UC-3 (Generate Violation – Voice)	AM-AN				UC3-TC02, UC3-TC03
RE-FR-07	UC-3 (Review Generated Violation)	AM-AN				UC3-TC04
RE-FR-08	UC-5 (Violation Management – Search/View)	BM-AN				UC5-TC01, UC5-TC02
RE-FR-09	UC-5 (Violation Management)	BM-AN				UC5-TC03
RE-FR-10	UC-5 (Search Violation)	BM-AN				UC5-TC04
RE-FR-11	UC-11 (View Violations – Citizen)	BM-AN				UC11-TC01
RE-FR-12	UC-12 (Submit Appeal)	BM-AN				UC12-TC01, UC12-TC02
RE-FR-13	UC-10 (Review Appeals)	BM-AN				UC10-TC01
RE-FR-14	UC-6 (Create Police Account)	BM-AN				UC6-TC01

RE-FR-15	UC-6 (Create Police Account)	BM-AN				UC6-TC02
RE-FR-16	UC-7 (Edit Police Account)	BM-AN				UC7-TC01
RE-FR-17	UC-8 (Activate / Deactivate Account)	BM-AN				UC8-TC01
RE-FR-18	UC-8 (Activate / Deactivate Account)	BM-AN				UC8-TC02
RE-FR-19	UC-9 (Review Violations)	VM-AN				UC9-TC01
RE-FR-20	UC-9 (Search Reviewed Violations)	VM-AN				UC9-TC02
RE-FR-21	UC-9 (View Violation Details)	VM-AN				UC9-TC03
RE-FR-22	UC-10 (Review Appeals)	VM-AN				UC10-TC02
RE-FR-23	UC-10 (Accept Appeal)	VM-AN				UC10-TC03
RE-FR-24	UC-10 (Reject Appeal)	VM-AN				UC10-TC04
RE-FR-25	UC-3 (Voice to Text)	VM-AN				UC3-TC05
RE-FR-26	UC-3 (Image Analysis – Plate Extraction)	VM-AN				UC3-TC06
RE-FR-27	UC-3 (Automated Report Generation)	VM-AN				UC3-TC07
RE-FR-28	UC-3 (Violation Classification)	VM-AN				UC3-TC08

Chapter 4

System Design

Introduction:

This chapter offers a comprehensive overview of the design phase steps, encompassing system design and architecture determination, as well as detailed design and modeling. It underscores the importance of these steps in creating a robust system for collecting, analyzing, and utilizing user feedback to continuously enhance the software system.

System Architecture:

in this section we will discuss system design where system design involves defining the overall structure and architecture of the software, ensuring that all components work seamlessly together. This phase establishes the blueprint for efficient data flow, interaction, and integration within the system

The first constraints to take into consideration when building an architecture for a software system are the **non-functional requirements (quality attributes)**, so first we will start with the non-functional requirements of our system:

1. Usability:

The system shall provide intuitive and user-friendly interfaces that enable traffic police officers, police managers, and citizens to perform their tasks efficiently with minimal training. The workflows shall be simple and optimized for field operations, supported by clear visual cues such as meaningful icons and guided input forms to reduce user errors.

2. Security

The system shall protect sensitive data, including user account information, traffic violation records, and appeal data, by implementing secure authentication mechanisms, role-based access control, and encrypted communication channels. All data exchanges shall be secured using HTTPS protocols, and access to system resources shall be restricted according to user roles.

3. Performance

The system shall ensure acceptable response times for interactive operations such as login, data retrieval, and search functionalities. AI-based processes, including image analysis and speech-to-text conversion, shall be executed asynchronously to prevent degradation of system responsiveness.

4. Reliability

The system shall ensure reliable operation and data integrity under normal and failure conditions. Mechanisms such as error handling, data validation, and safe data persistence shall be implemented to prevent data loss during system or network failures.

5. Availability

The system shall be available during operational hours with minimal downtime. The design shall support continuous system operation to accommodate multiple concurrent users, including traffic police officers operating in the field.

Components functionalities:

API Gateway:

Routes incoming API requests from client applications to the appropriate backend services and enforces access control.

User Management:

Manages user authentication, authorization, and account-related operations.

Violation Management:

Handles the creation, storage, retrieval, and management of traffic violation records.

Speech-to-Text:

Converts voice inputs from police officers into text for violation processing.

Plate Recognition:

Extracts vehicle plate numbers and related information from captured images.

Violation Generation:

Generates structured traffic violation reports by combining AI outputs and user input.

Data Access:

Provides standardized communication between system services and databases.

SQL Database:

Stores structured system data such as user accounts and violation records.

NoSQL Database:

Stores unstructured data generated by AI services, such as audio and image analysis results.

System Architecture:

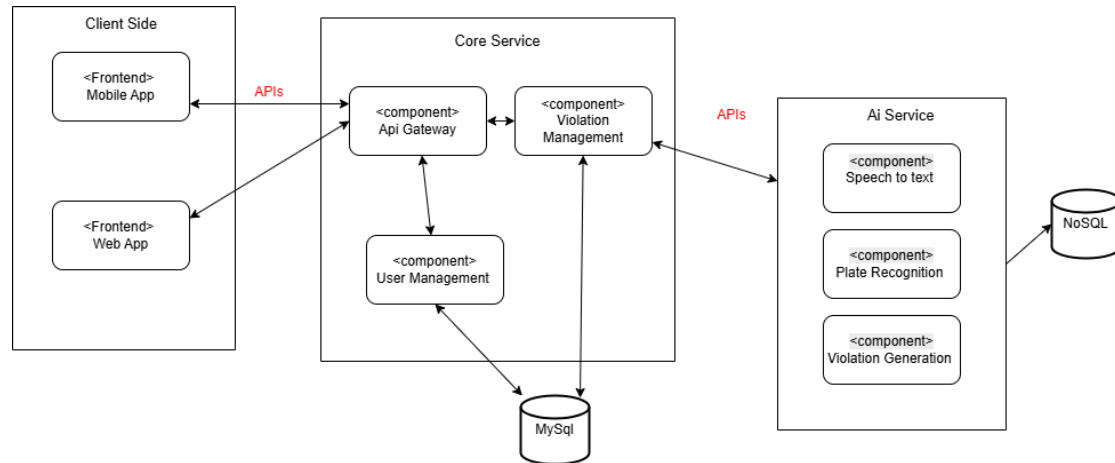


Figure 18 system architecture

The **Smart Traffic Police Assistant System** is designed using a **microservices architecture** combined with a **client-server model**, where system functionalities are divided into independent and loosely coupled services to improve scalability, maintainability, and fault isolation.

The **client side** consists of a mobile application and a web application that communicate with the backend through secured RESTful APIs.

The **backend** is composed of multiple independent microservices grouped into two main layers:

Core Services:

This layer includes the **API Gateway**, **User Management**, and **Violation Management** services.

The API Gateway acts as a single entry point for all client requests and routes them to the appropriate microservices. The User Management service handles authentication and user account operations, while the Violation Management service manages traffic violation records and related operations.

AI Services:

This layer contains specialized microservices responsible for advanced processing tasks, including **Speech-to-Text**, **Plate Recognition**, and **Violation Generation**. These services operate independently and communicate with the core services via APIs.

Each microservice manages its own data access. Structured system data such as user accounts and violation records are stored in a **MySQL database**, while AI-generated and unstructured data are stored in a **NoSQL databas**

Database Design

SQL Database – MySQL:

In the **Smart Traffic Police Assistant System**, **MySQL** is used as the primary relational database to manage highly structured and transactional system data. This includes user accounts, traffic violation records, appeal data, roles, permissions, and system session information. MySQL provides strong data consistency, referential integrity, and efficient support for complex queries required for administrative, monitoring, and reporting functionalities.

The database schema is designed following normalization principles to reduce redundancy and enforce clear relationships between core entities, including:

users :

Stores user account information, including authentication credentials, role type (police officer, police manager, administrator, citizen), and account status.

violations:

Contains traffic violation records, including violation type, time, location, vehicle plate number, and status.

appeals:

Stores citizen appeal information linked to specific violations, including appeal reason, submission date, and decision status.

roles and permissions:

Manages role-based access control to ensure that system functionalities are accessed according to user privileges.

sessions and system logs:

Maintains session tracking, authentication states, and system activity logs for auditing and security purposes.

Foreign key constraints are applied to enforce data integrity across the system, such as linking **user_id** in the violations table to the users table, and **violation_id** in the appeals table to the violations table. This ensures consistent and reliable relationships between users, violations, and appeals.

MySQL- database modeling:

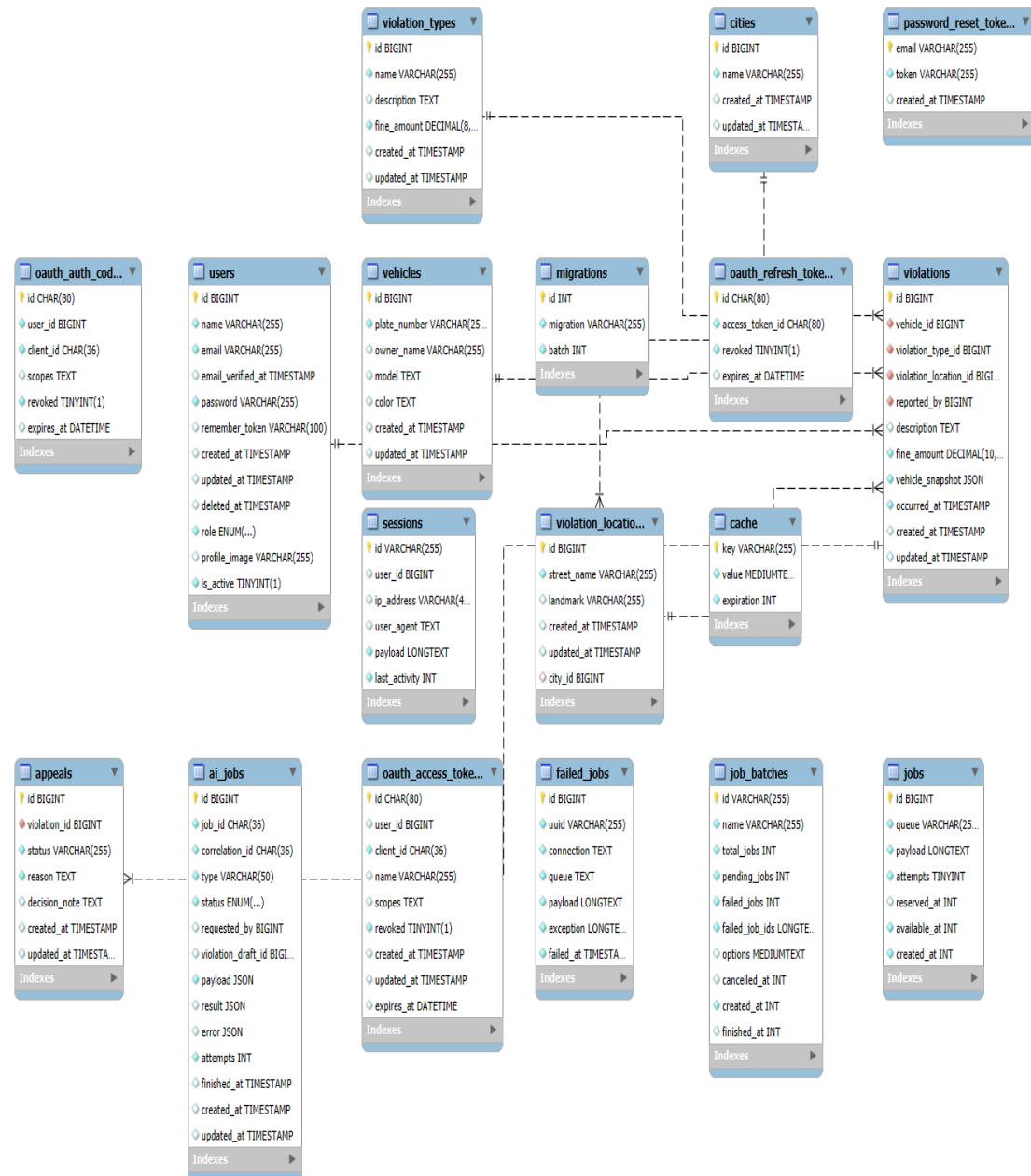


Figure 19 "MY_SQL"

Updating the RTM:

Req-ID	Use Case	Analysis	System Design	Detailed Design	Coding	Test Case(s)
RE-FR-01	UC-1	AM-AN	User Management Service	—		UC1-TC01, UC1-TC02
RE-FR-02	UC-1	AM-AN	User Management Service	—		UC1-TC03
RE-FR-03	UC-2	AM-AN	User Management Service	—		UC2-TC01, UC2-TC02
RE-FR-04	UC-3	AM-AN	Violation + AI Services	—		UC3-TC01
RE-FR-05	UC-4	AM-AN	Violation Management	—		UC4-TC01, UC4-TC02
RE-FR-06	UC-3	AM-AN	Plate Recognition Service	—		UC3-TC02, UC3-TC03
RE-FR-07	UC-3	AM-AN	Violation Management	—		UC3-TC04
RE-FR-08	UC-3	AM-AN	Violation Management	—		UC3-TC05
RE-FR-09	UC-5	BM-AN	Violation Management	—		UC5-TC01
RE-FR-10	UC-5	BM-AN	Violation Management	—		UC5-TC02
RE-FR-11	UC-11	BM-AN	Violation Management	—		UC11-TC01
RE-FR-12	UC-12	BM-AN	Appeal Management	—		UC12-TC01

RE-FR-13	UC-12	BM-AN	Appeal Management	—		UC12-TC02
RE-FR-14	UC-1	AM-AN	User Management Service	—		UC1-TC01
RE-FR-15	UC-1	AM-AN	User Management Service	—		UC1-TC03
RE-FR-16	UC-2	AM-AN	User Management Service	—		UC2-TC01
RE-FR-17	UC-6	BM-AN	User Management Service	—		UC6-TC01
RE-FR-18	UC-7	BM-AN	User Management Service	—		UC7-TC01
RE-FR-19	UC-7	BM-AN	User Management Service	—		UC7-TC02
RE-FR-20	UC-8	BM-AN	User Management Service	—		UC8-TC01
RE-FR-21	UC-6	BM-AN	User Management Service	—		UC6-TC02
RE-FR-22	UC-10	DM-AN	Appeal Management	—		UC10-TC01
RE-FR-23	UC-10	DM-AN	Appeal Management	—		UC10-TC02
RE-FR-24	UC-10	DM-AN	Appeal Management	—		UC10-TC03
RE-FR-25	UC-3	FGM-AN	AI – Speech to Text	—		UC3-TC06
RE-FR-26	UC-3	FGM-AN	AI – Plate Recognition	—		UC3-TC07
RE-FR-27	UC-3	FGM-AN	AI – Violation Generation	—		UC3-TC08

RE-FR-28	UC-3	FGM-AN	AI – Classification	—		UC3-TC09
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Chapter 5

AI service

1. Introduction

Artificial Intelligence (AI) is playing an increasingly important role in modern information systems by automating complex tasks, improving data accuracy, and enhancing decision-making processes. In the domain of traffic management and law enforcement, traffic police officers are required to document violations accurately while operating in dynamic and often challenging field environments. Manual data entry during traffic enforcement is time-consuming, error-prone, and can distract officers from their primary duties.

The AI module in the Traffic Police Assistant System is designed to address these challenges by automating the extraction of critical information from unstructured inputs such as spoken audio and captured images. By leveraging advanced Speech-to-Text (STT) and computer vision technologies, the system enables traffic police officers to record violations efficiently and accurately without relying on manual typing.

Designed as an independent and modular component, the AI service enhances the overall efficiency, accuracy, and reliability of traffic violation documentation, supporting smarter and faster law enforcement workflows.

2. Key Functionalities of the AI Module

The AI module processes traffic violation data through multiple intelligent components, each responsible for handling a specific type of input:

- **Speech-to-Text (STT):**
Converts the spoken description of traffic violations recorded by the police officer into structured textual data. The STT component supports the Arabic language and is designed to handle dialectal variations and real-world noise conditions.
- **License Plate Recognition (Vision & OCR):**
Extracts vehicle license plate numbers from captured images using computer vision and text recognition models. This component ensures accurate identification of vehicles involved in traffic violations.
- **Data Structuring and Validation:**
Organizes the extracted textual information into structured fields such as violation type, vehicle details, location, and additional notes, enabling efficient storage and further processing within the system.

In this chapter, we present the core AI technologies used in the Traffic Police Assistant System, review related work and existing models, and describe the design, implementation, and experimental evaluation of the proposed AI components.

3. Basic Concepts :

1. General Concept of Artificial Intelligence in the System

The Traffic Police Assistant System relies on Artificial Intelligence (AI) techniques to automate the processing of unstructured data, such as audio and images, generated during field operations. The main objective of the AI component is to transform this unstructured input into structured and usable textual information that can support traffic violation documentation and decision-making.

Conceptually, AI in this system aims to replicate selected human capabilities, such as hearing and visual perception. The system is designed to understand what the traffic police officer says and analyze what the camera captures, without requiring extensive manual data entry.

2. Core AI Components of the System

From a high-level perspective, the AI module of the proposed system is divided into two main components that operate independently but complement each other:

Speech Processing Component

This component focuses on converting the spoken description of a traffic violation recorded by the police officer into written text using Speech-to-Text (STT) technologies. This approach enables faster documentation, reduces manual effort, and allows hands-free interaction in real-world traffic environments.

Vision-Based Processing Component

This component is responsible for extracting visual information from captured images, particularly vehicle license plate numbers. It relies on computer vision techniques and Optical Character Recognition (OCR) or vision-language models to analyze images and extract textual information accurately.

3. Speech-to-Text Concept

Speech-to-Text is an AI application that converts human speech signals into corresponding text. This process involves analyzing audio signals, extracting

acoustic features, and mapping them to linguistic representations using machine learning and deep learning models.

In the context of the Traffic Police Assistant System, STT is used to convert spoken traffic violation descriptions into official textual records. However, Arabic speech recognition introduces additional challenges, such as dialectal variations, pronunciation differences, and environmental noise, which must be carefully considered when selecting an appropriate STT model.

4. Computer Vision and Text Extraction Concept

Computer vision refers to the ability of intelligent systems to interpret and analyze visual content. In this project, vision-based techniques are employed to extract license plate numbers from vehicle images captured at the scene of the violation.

This process can be implemented using a traditional two-stage pipeline, consisting of license plate detection followed by OCR, or through modern vision-language models that integrate detection and text extraction within a single framework.

5. Integration of AI Components

The integration of speech-based and vision-based AI components allows the Traffic Police Assistant System to generate a comprehensive and reliable traffic violation record. By combining descriptive information obtained from speech with objective visual data extracted from images, the system improves data accuracy, reduces human error, and enhances overall operational efficiency.

4. Literature review:

AI Literature Review

License Plate Recognition and OCR Models

1. Introduction

Vehicle License Plate Recognition (LPR) is a critical component in intelligent traffic enforcement systems. The task aims to automatically identify vehicle license numbers from images captured in real-world environments. From a conceptual perspective, the LPR process consists of two main stages: detecting the license plate region within an image and extracting the alphanumeric characters from that region.

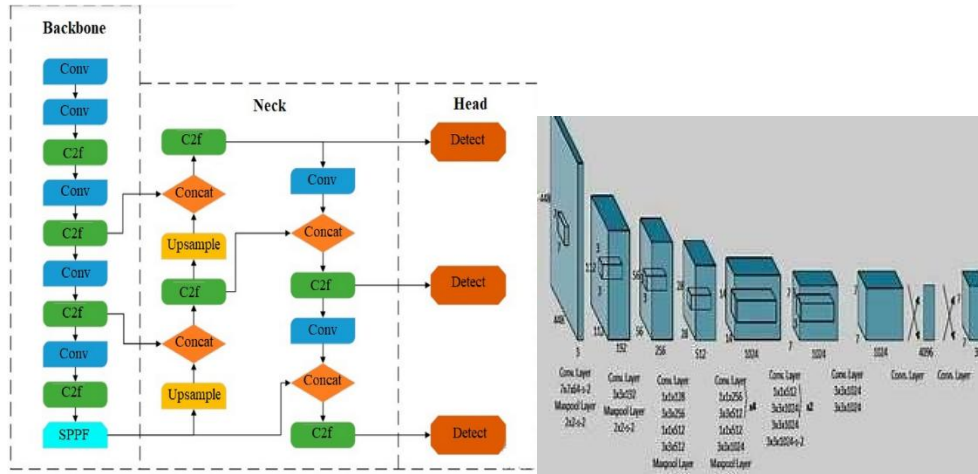
Traditionally, this task has been addressed using a two-stage pipeline that combines object detection models with Optical Character Recognition (OCR) techniques. While effective in controlled scenarios, this approach increases system complexity and introduces error propagation between detection and recognition stages.

Recent advances in vision-language models have introduced an alternative unified approach, where detection and text extraction are handled within a single model. This study evaluates both traditional and modern approaches to identify the most suitable solution for the Traffic Police Assistant System.

2. Traditional License Plate Recognition Pipeline

2.1 YOLOv8 for License Plate Detection

YOLOv8 is a real-time object detection model widely used for detecting objects such as vehicles and license plates. In license plate recognition systems, YOLOv8 is responsible for localizing the license plate region, which is then passed to an OCR model for character extraction.



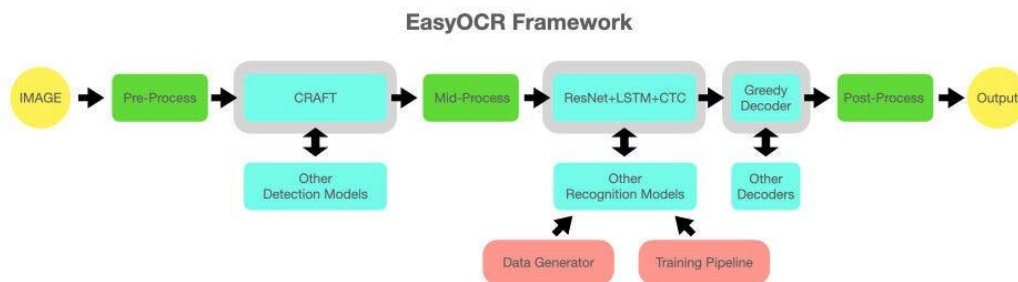
Although YOLOv8 provides fast and accurate detection, the overall system performance depends heavily on the accuracy of the subsequent OCR stage.

2.2 OCR Models for Text Extraction

Once the license plate is detected, OCR models are used to extract the alphanumeric characters. In this study, EasyOCR and Tesseract OCR are evaluated as traditional OCR solutions.

2.2.1 EasyOCR

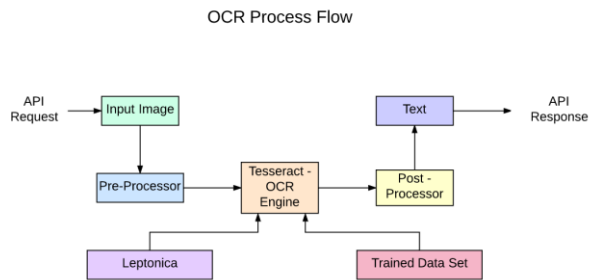
EasyOCR is an open-source OCR framework based on deep learning architectures. It supports multiple languages and is widely used for general OCR tasks.



However, EasyOCR struggles with license plate images due to small character size, varying lighting conditions, and mixed Arabic–English digits, often resulting in partial or incorrect recognition.

2.2.2 Tesseract OCR

Tesseract OCR is one of the most widely adopted open-source OCR engines, originally developed for document-style text recognition.

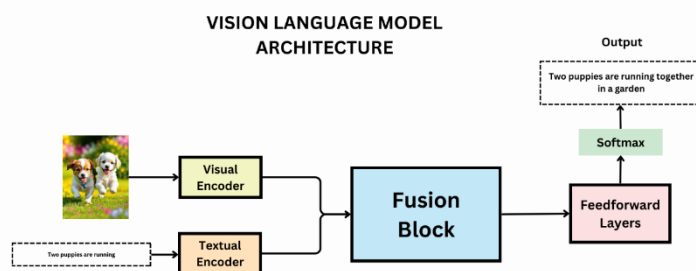


Despite its success in document processing, Tesseract performs poorly in scene text recognition tasks such as license plates, where noise, perspective distortion, and complex backgrounds are present.

3. Vision-Language Model Approach

3.1 Qwen3 VL:2b

Qwen3 VL:2b is a modern vision-language model designed to jointly process visual and textual information. Unlike traditional pipelines, it can directly generate textual output from an image without requiring a separate detection or OCR stage.



This unified architecture significantly reduces pipeline complexity and minimizes error propagation, making Qwen3 VL:2b particularly suitable for license plate recognition in complex real-world scenarios.

4. Experimental Evaluation

4.1 EasyOCR (Without Detection)



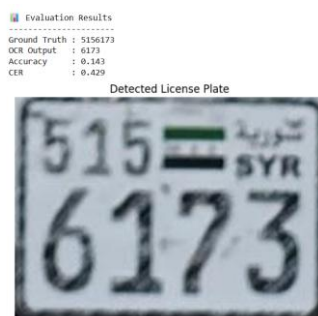
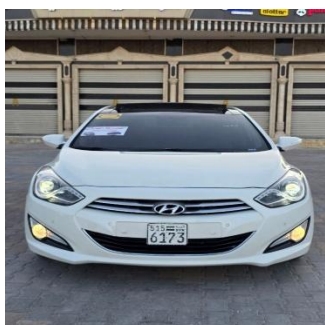
Ground Truth OCR Output Accuracy CER

5108703 5855103 0.429 0.571

The results indicate that EasyOCR alone fails to extract complete license plate numbers accurately.

4.2 EasyOCR with YOLOv8

Combining EasyOCR with YOLOv8 improves plate localization but does not resolve incomplete character recognition. In many cases, only part of the license plate number is extracted.



4.3 Qwen3 VL:2b Results



Example Ground Truth Output Accuracy CER

1	5077237	5077237	1.00	0.0
2	5103682	5103682	1.00	0.0
3	233622	233622	1.00	0.0

Qwen3 VL:2b achieved perfect recognition across all tested cases.

4.4 Tesseract OCR Results

Tesseract OCR failed to produce reliable results both independently and when combined with YOLOv8, confirming its limitations in real-world license plate recognition.

5. Summary of Key Findings

EasyOCR produced high character error rates and incomplete outputs.

EasyOCR combined with YOLOv8 improved localization but not recognition accuracy.

Tesseract OCR was unsuitable for scene text recognition tasks.

Qwen3 VL:2b achieved superior accuracy and zero CER while simplifying the pipeline.

6. Conclusion

Based on experimental results, Qwen3 VL:2b was selected as the primary license plate recognition model for the Traffic Police Assistant System due to its high accuracy, robustness, and integrated architecture.

Speech-to-Text (STT) Models

1. Introduction

Speech-to-Text (STT) technology plays a critical role in intelligent systems that rely on spoken input to generate structured textual data. In traffic enforcement environments, police officers often operate under time pressure and challenging conditions, making manual text entry inefficient and error-prone. STT provides an effective solution by enabling officers to verbally describe traffic violations while the system automatically converts speech into text.

Within the Traffic Police Assistant System, the STT component is responsible for transforming spoken violation descriptions—including vehicle details, locations, and

violation types—into accurate textual records. Due to the linguistic complexity of Arabic, environmental noise, and dialectal variations, selecting a robust and reliable STT model is essential.

This study presents a comparative evaluation of state-of-the-art STT models, focusing on their suitability for real-world deployment in traffic enforcement scenarios.

2. Overview of Speech-to-Text Technology

Speech-to-Text systems aim to convert human speech signals into written text by analyzing acoustic patterns and mapping them to linguistic representations. Modern STT systems are primarily based on deep learning architectures, particularly Transformer models, which have demonstrated strong performance in multilingual and noisy environments.

An STT pipeline typically involves:

Audio preprocessing and feature extraction

Acoustic modeling

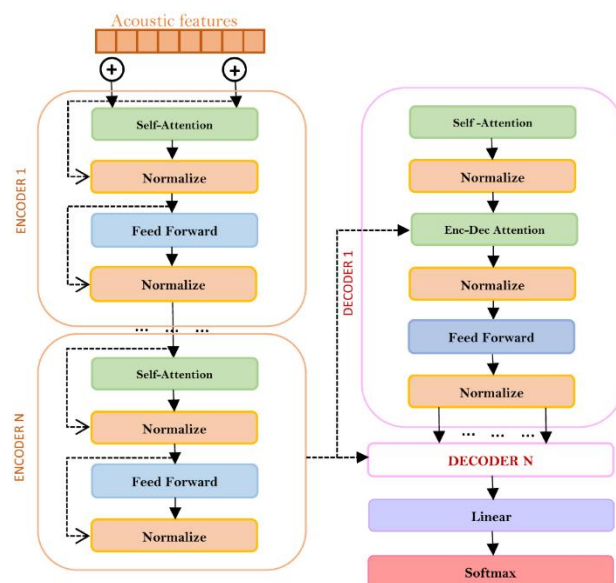
Language modeling

Text decoding

3.1 Whisper

Whisper is an automatic speech recognition system developed by OpenAI and trained on approximately 680,000 hours of multilingual and multitask supervised data collected from the web. This large-scale and diverse dataset enables Whisper to achieve high robustness against accents, background noise, and technical vocabulary.

Whisper follows an end-to-end encoder–decoder Transformer architecture. Input audio is segmented into 30-second chunks, converted into log-Mel spectrograms, and passed through the encoder. The decoder then generates the corresponding text output, guided by special tokens that support tasks such as language identification, transcription, and translation.



Whisper is available in multiple sizes (tiny, base, small, medium, large), allowing trade-offs between accuracy, processing speed, and memory usage.

3.2 Vosk (Arabic-Supported STT)

Vosk is an open-source offline speech recognition toolkit designed for lightweight and real-time applications. It supports multiple languages, including Arabic, and can operate without an internet connection, making it suitable for privacy-sensitive environments.

Vosk relies on traditional acoustic and language modeling approaches optimized for low resource consumption. While it offers fast inference and low memory usage, its recognition accuracy—particularly for Arabic dialects—is often lower than modern Transformer-based models.

4. Comparison Criteria

To objectively evaluate the STT models, the following comparison criteria were defined prior to experimentation:

Word Error Rate (WER): Measures transcription errors at the word level

Accuracy Percentage: Calculated as $(1 - \text{WER}) \times 100$

Processing Time: Time required to transcribe a single audio file

Memory Consumption (RAM): Runtime memory usage

Language Support: Ability to accurately transcribe Arabic speech

Offline Capability: Suitability for deployment without internet access

Robustness: Performance under noisy and real-world conditions

5. Experimental Setup

All experiments were conducted in a local offline environment to simulate real-world deployment conditions for traffic police officers.

Experimental setup included:

Audio recordings of traffic violation descriptions

Arabic language recordings by native speakers

Realistic background noise conditions

Ground truth transcripts manually prepared for evaluation

WER computation using standard evaluation tools

Two Whisper model variants (Small and Medium) were evaluated and compared with Vosk for Arabic speech recognition.

6. Experimental Results

6.1 Whisper – Small Model

Processing Time: ~23 seconds per audio

Accuracy: ~30%

Observations:

Fast execution

Low memory usage

High transcription errors in Arabic

6.2 Whisper – Medium Model

Processing Time: ~105 seconds per audio

Accuracy: ~70%

Observations:

Significantly improved accuracy

Better handling of Arabic pronunciation

Higher memory consumption

6.3 Vosk (Arabic)

Processing Time: Very fast

Accuracy: Low

Observations:

Supports Arabic offline

Fails to capture full semantic meaning

Not suitable for detailed violation descriptions

7. Discussion

The experimental results demonstrate a clear trade-off between model size, accuracy, and computational cost. While Whisper Small offers faster inference, its transcription accuracy for Arabic speech is insufficient for reliable traffic violation documentation. Whisper Medium, despite higher processing time and memory requirements, provides a substantially better transcription quality.

Vosk, although lightweight and offline-capable, fails to meet the accuracy requirements necessary for legal and administrative traffic enforcement tasks.

The superior performance of Whisper Medium can be attributed to its larger model capacity and extensive multilingual training data, which enable better generalization across Arabic speech patterns.

8. Conclusion

Based on the comparative evaluation, Whisper Medium was selected as the primary Speech-to-Text model for the Traffic Police Assistant System. It provides the best balance between transcription accuracy and robustness for Arabic speech, despite higher computational requirements.

Future work may explore optimization techniques, model quantization, or GPU acceleration to improve inference speed while maintaining accuracy.

Service Design and Implementation

3.1 Introduction

In this section, we present the design and implementation of the Artificial Intelligence (AI) service used in the Traffic Police Assistant System. The AI service is designed to efficiently process unstructured inputs generated during traffic enforcement operations, particularly audio recordings and captured images, and transform them into structured and reliable traffic violation records.

The service follows a modular and scalable architecture that enables independent processing of Speech-to-Text (STT) and License Plate Recognition components. By

leveraging advanced deep learning models and well-defined processing stages, the AI service supports accurate transcription of spoken violation descriptions, extraction of vehicle information from images, and organization of the extracted data into a structured format suitable for storage and further analysis.

3.2 AI Service Design

The AI service is designed as an independent processing component that can be integrated with external systems through well-defined APIs. Its architecture supports asynchronous processing, allowing traffic officers to submit violation data without waiting for immediate results, while the system processes inputs in the background.

The service pipeline consists of multiple stages, each responsible for a specific task, including audio transcription, image-based license plate recognition, and data structuring. This design ensures flexibility, maintainability, and ease of future expansion.

3.2.1 Service Functional Requirements

The functional requirements of the AI service are defined as follows:

The service must support receiving audio recordings of traffic violation descriptions recorded by traffic police officers.

The service must support receiving vehicle images captured during traffic enforcement for license plate recognition.

The service must be capable of retrieving audio and image files from external storage sources (e.g., cloud storage or local repositories) for processing.

The service must convert received audio files into accurate textual transcripts using the Speech-to-Text (STT) component.

The service must support Arabic language transcription and handle realistic field conditions, including background noise and pronunciation variations.

The service must extract vehicle license plate numbers from captured images using computer vision and OCR or vision-language models.

The service must organize the extracted information into structured fields, including violation description, vehicle details, location, and additional notes.

The service must provide an API interface that allows external systems or applications to submit data for processing.

The service must track and store the processing status of each request at different stages (e.g., STT processing, image analysis, data structuring).

The service must provide an API endpoint to retrieve the real-time processing status of submitted requests.

The service must provide an API endpoint to retrieve the final structured traffic violation record, including transcribed text and extracted license plate information.

Non-Functional Requirements :

1) Performance

- The AI Service must process jobs asynchronously; STT/OCR processing time depends on file size but must be measured and monitored.
- The system must enforce file size limits (e.g., 25–50MB) and return clear validation messages.

2) Scalability

The system must support horizontal scaling by increasing RabbitMQ consumers.
The AI Service should allow multiple instances/workers (STT/OCR) without conflicts.

3) Reliability

- No job loss: every job published to RabbitMQ must be processed or retried safely.
Implement a retry policy with a maximum attempt count and a Dead Letter Queue (DLQ) for failures.
- Persist job states (Queued / Processing / Done / Failed) in metadata storage.

Processing Flow:

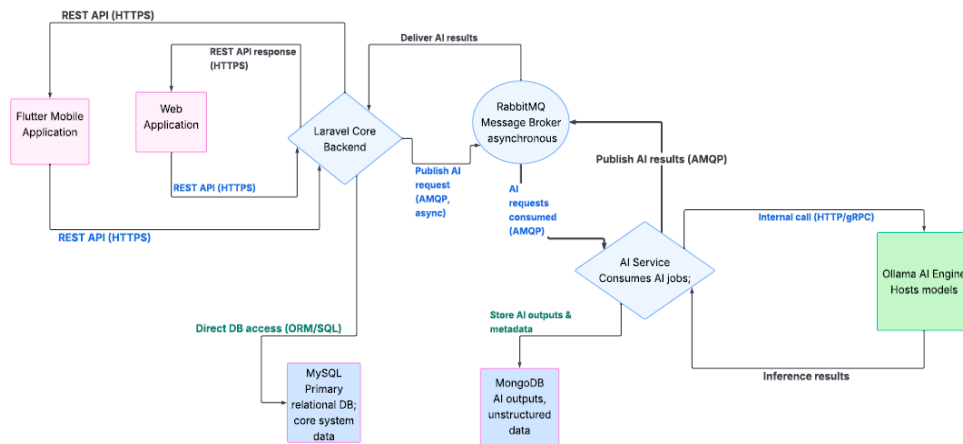


Figure : AI Service Process Flow in the Traffic Police Assistant System

This figure illustrates the end-to-end processing flow. Client applications (Flutter/Web) submit requests to the Laravel backend via REST APIs. The backend publishes AI jobs (STT/OCR) asynchronously to RabbitMQ. The AI Service consumes these jobs, calls the model engine (Ollama) for inference, and stores AI outputs and metadata in MongoDB, while core system data remains in MySQL. Finally, Laravel retrieves the results and exposes them back to clients through API endpoints.

AI Processing Steps

The AI service processes traffic violation data through a structured multi-stage pipeline. The processing steps are as follows:

- Audio Acquisition:**
 Receive recorded audio files from the mobile application or retrieve them from external storage through secure API endpoints.
- Audio Preprocessing:**
 Normalize the audio format, sampling rate, and channels to ensure compatibility with the Speech-to-Text model.
- Speech-to-Text Conversion:**
 Convert the processed audio input into textual form using the selected Speech-to-Text model, producing a raw transcription of the traffic violation description.
- Text Post-processing:**
 Clean and normalize the generated text by correcting spacing, removing noise tokens, and standardizing numbers and entities (e.g., vehicle numbers, locations).
- Image Acquisition (OCR Path):**
 Receive vehicle images captured by the traffic officer or retrieved from storage for license plate recognition.

- **License Plate Recognition:**
Extract the license plate number either through a traditional detection and OCR pipeline (YOLOv8 + OCR) or through a unified vision-language model.
- **Result Validation:**
Validate the extracted textual outputs by checking confidence scores, format consistency, and language correctness.
- **Data Storage:**
Store structured AI outputs and metadata in the database for later retrieval and auditing.
- **Result Delivery:**
Return the final transcription and recognition results to the backend service through asynchronous message communication.

Used Tools and Models

This section presents the main tools, frameworks, and artificial intelligence models used in the development and implementation of the AI service within the Traffic Police Assistant System. These tools were selected based on their reliability, performance, and suitability for real-world traffic enforcement scenarios.

1. Backend and Integration Tools

Laravel Framework

Laravel is used as the core backend framework responsible for handling API requests, managing authentication, orchestrating AI processing jobs, and communicating with other system components. It provides a clean architecture, robust security features, and seamless integration with databases and message queues.

RESTful APIs (HTTPS)

REST APIs are used to enable communication between client applications (Flutter and Web) and the backend services. All APIs use HTTPS to ensure secure data transmission.

RabbitMQ (Message Broker)

RabbitMQ is used as an asynchronous message broker to manage AI processing jobs. It decouples the backend from the AI service, allowing scalable and fault-tolerant job processing. Each AI task (STT or OCR) is published as a message and consumed by the AI service workers.

2. Databases

MongoDB

MongoDB is used to store unstructured AI outputs and metadata, including transcription

results, OCR outputs, confidence scores, and processing logs. Its flexible schema makes it suitable for handling AI-generated data.

3. Speech-to-Text (STT) Models and Tools

Whisper

Whisper is used as the primary Speech-to-Text model. The Medium version was selected due to its superior accuracy in Arabic speech recognition compared to smaller variants. Whisper converts spoken traffic violation descriptions into structured text.

Python

Python is used to implement the STT processing logic and model execution due to its strong ecosystem for machine learning and audio processing.

4. License Plate Recognition and OCR Models

Qwen3 VL:2b

Qwen3 VL:2b is the primary vision-language model used for license plate recognition. It performs image-to-text extraction directly, eliminating the need for a separate detection and OCR pipeline.

5. AI Inference Engine

Ollama AI Engine

Ollama is used to host and run AI models locally, including vision-language models. It enables on-premise inference, enhances data privacy, and reduces dependency on external cloud services.

6. Frontend Tools

Flutter Mobile Application

Flutter is used to develop the mobile application for traffic police officers, allowing them to record audio, capture images, and submit violations.

Web Application

A web-based interface is provided for administrative review, monitoring AI processing status, and accessing structured violation reports.

Chapter 6

Practical Implementation

Introduction:

In this chapter, we will present the practical implementation of the system, detailing the technologies and tools employed. Additionally, we will showcase the

system interfaces and conclude by executing test cases to verify the system's functionalities across different scenarios.

Used Tools:

- **Laravel Framework**

Laravel is a powerful PHP web framework designed to simplify the development of robust and scalable web applications. It follows the MVC (Model–View–Controller) architecture and provides built-in features such as routing, authentication, validation, and database abstraction. In our system, Laravel is used as the core backend framework responsible for handling RESTful APIs, managing user requests, orchestrating AI processing workflows, and integrating with databases and message queues.

- **Flutter**

Flutter is an open-source UI framework developed by Google for building cross-platform applications from a single codebase. It enables the development of high-performance mobile applications with a rich and responsive user interface. In this project, Flutter is used to build the mobile application used by traffic police officers to record audio, capture vehicle images, and submit traffic violations efficiently.

- **RabbitMQ**

RabbitMQ is an open-source message broker that implements asynchronous message queuing. It enables reliable communication between distributed system components. In our system, RabbitMQ is used to manage AI processing jobs asynchronously, allowing the backend to send Speech-to-Text and OCR requests to the AI service without blocking, thus improving scalability and system performance.

- **Python**

Python is a high-level programming language widely used in artificial intelligence and machine learning applications. It offers a rich ecosystem of libraries and frameworks for audio processing, computer vision, and natural language processing. In this project, Python is used to implement the AI service, including Speech-to-Text processing and OCR-related tasks.

- **Whisper (Speech-to-Text Model)**

Whisper is an automatic speech recognition (ASR) model developed by OpenAI. It is trained on large-scale multilingual datasets and demonstrates strong robustness to noise, accents, and different speaking styles. The Whisper Medium model is used in this system to convert spoken traffic violation descriptions into accurate textual transcripts, with strong support for the Arabic language.

- **Qwen3 VL:2b**

Qwen3 VL:2b is a modern vision-language model capable of processing images and generating textual outputs directly. Unlike traditional OCR pipelines, it can perform image-to-text recognition without requiring a separate object detection stage. In this project, Qwen3 VL:2b is used as the primary model for license plate recognition due to its high accuracy and simplified processing workflow.

- Ollama AI Engine

Ollama is a local AI inference engine used to host and run large language and vision-language models on-premise. It enables efficient and privacy-preserving execution of AI models without relying on external cloud services. In this system, Ollama is used to run vision-language models for OCR tasks.

- MySQL Database

MySQL is an open-source relational database management system widely used for structured data storage. It is known for its reliability, scalability, and performance. In this project, MySQL is used to store core system data such as users, vehicles, traffic violations, and AI processing references.

- MongoDB

MongoDB is a NoSQL database designed for storing unstructured and semi-structured data. It provides flexible schemas and high scalability. In this system, MongoDB is used to store AI-generated outputs, including STT transcripts, OCR results, confidence scores, and processing metadata.

- Postman

Postman is an API testing and development tool that simplifies the process of designing, testing, and debugging RESTful APIs. It is used extensively during development to validate backend endpoints and AI service integrations.

- Visual Studio Code (VS Code)

Visual Studio Code is a lightweight yet powerful source code editor that supports multiple programming languages and development tools. It provides features such as intelligent code completion, debugging, and extension support. In this project, VS Code is used to develop backend services, AI modules, and integration logic.

System Interfaces:

Police Officer interfaces:

login interface:

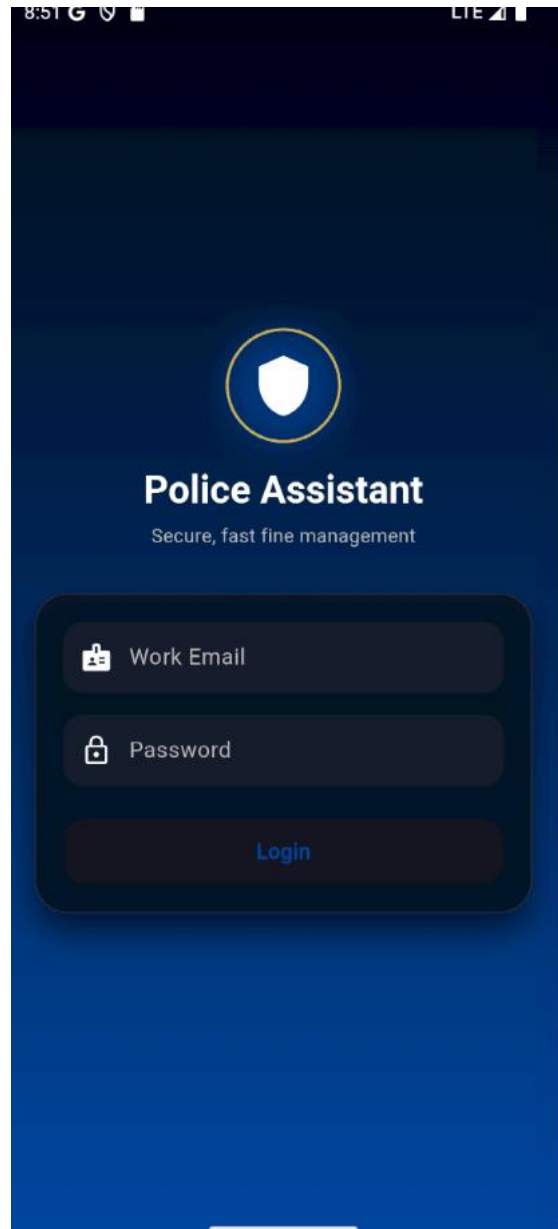
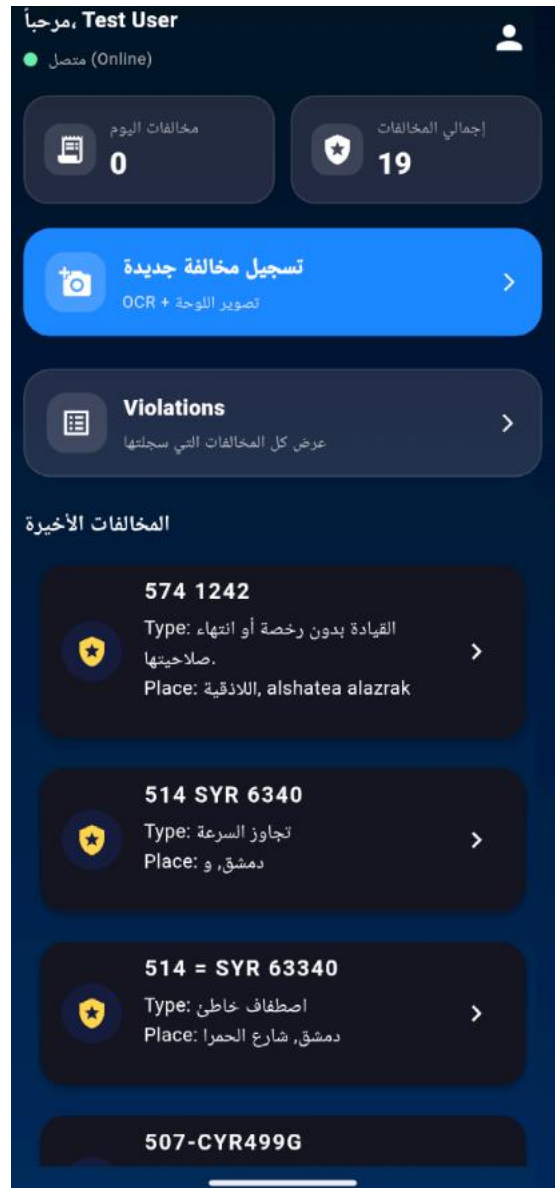


Figure login interface

homepage interface:



View Violation interface



View Violation details interface



Vehicle Information

Owner
salam alrashed

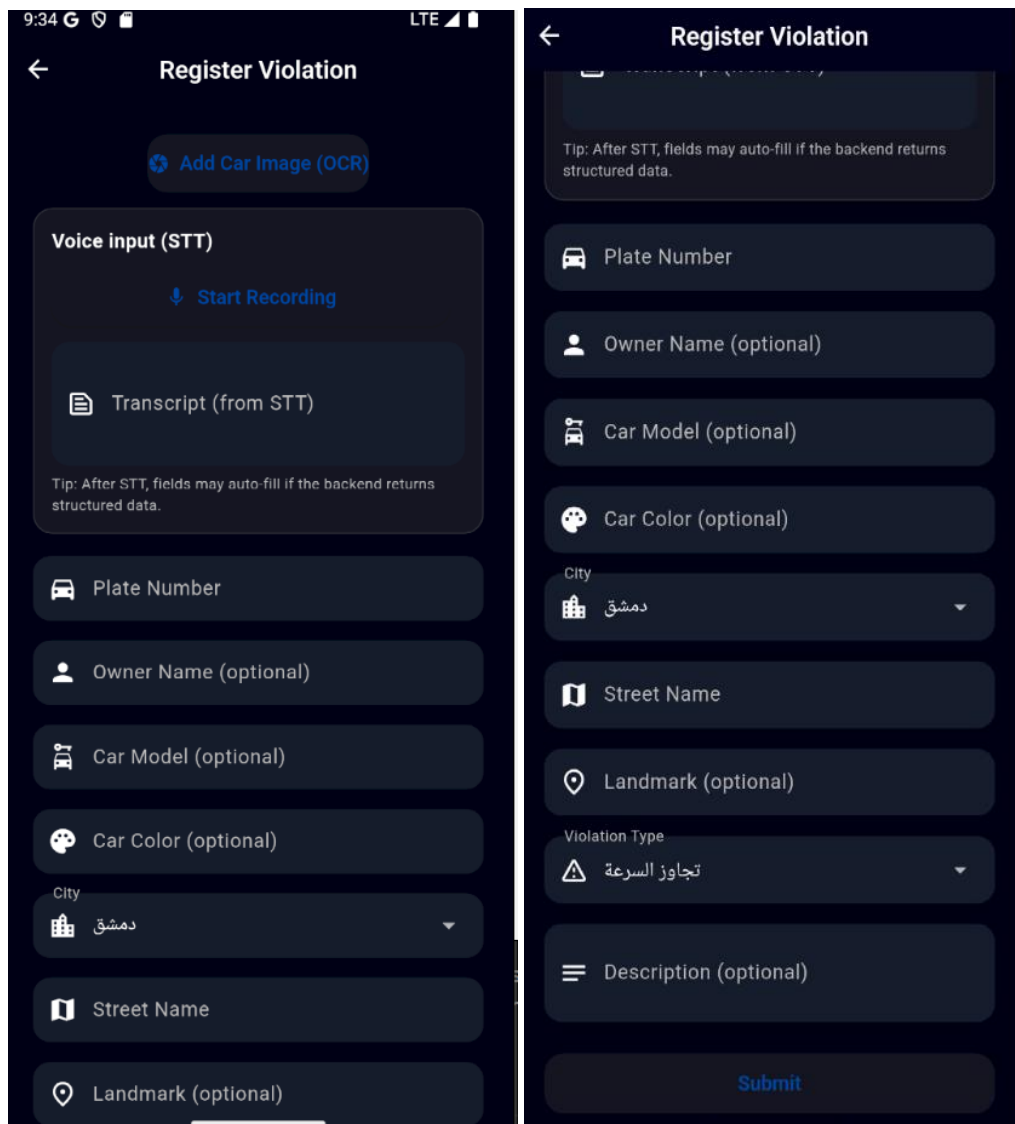
Street
alshatea alazrak

Violation Details

100 Fine Amount
100000

 Date

90



Profile settings interface:

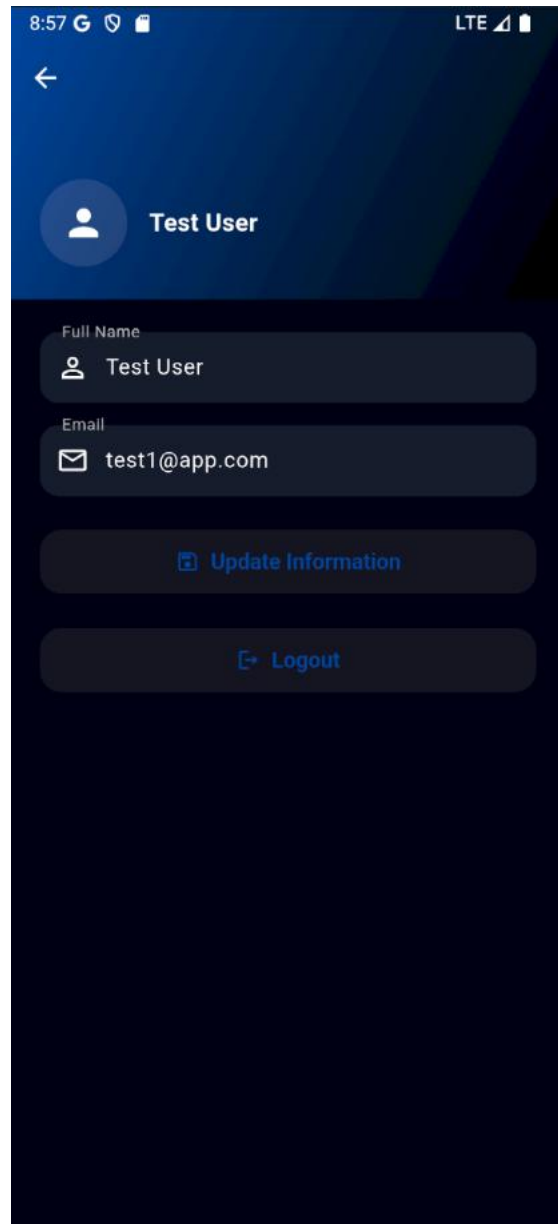
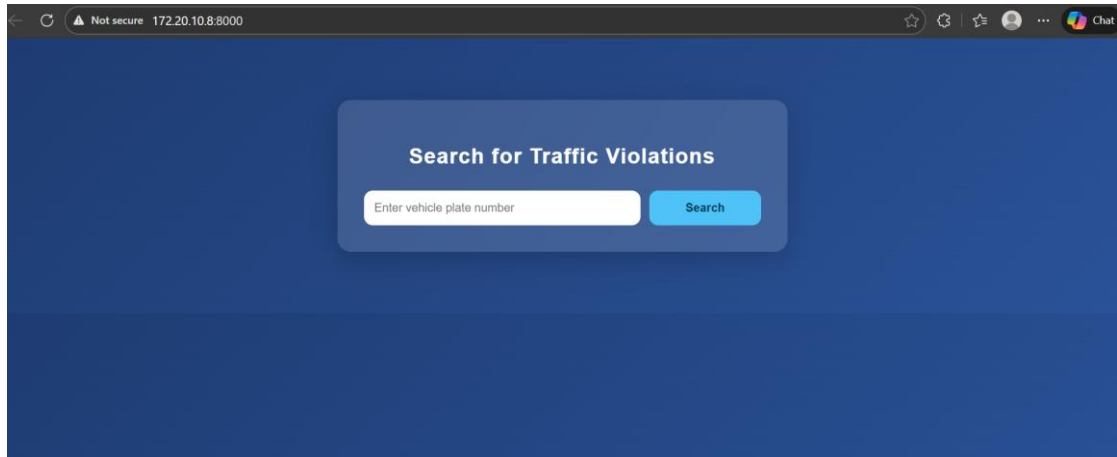


Figure 25 profile settings interface

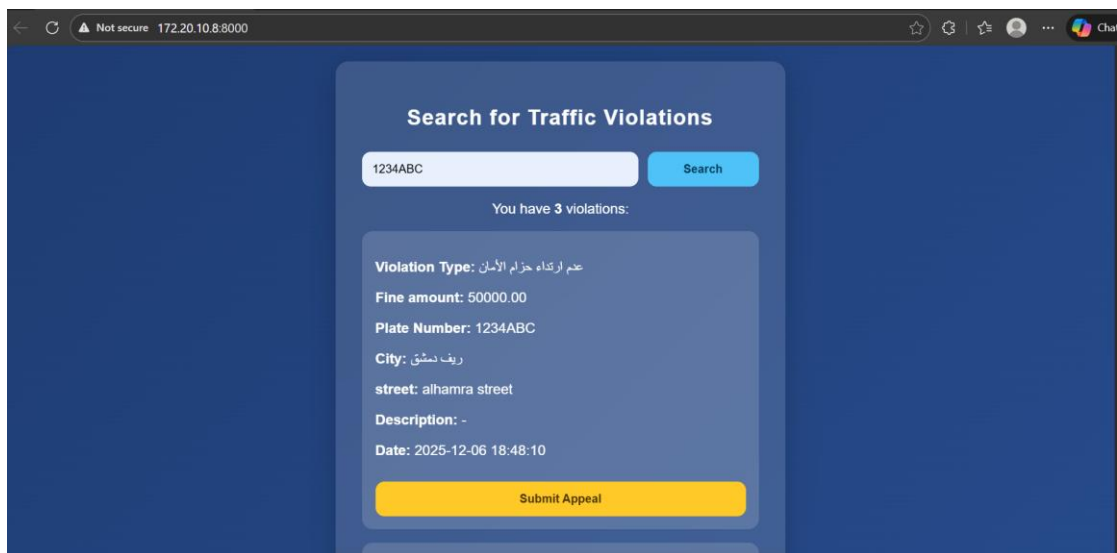
Citizen interfaces:

Search for Violation



The screenshot shows a web browser window with the address bar displaying 'Not secure 172.20.10.8:8000'. The page has a dark blue background. In the center, there is a light blue rounded rectangle containing the title 'Search for Traffic Violations'. Below the title is a white input field with the placeholder text 'Enter vehicle plate number' and a blue 'Search' button to its right.

View violation list



The screenshot shows the same web application after a search. The input field now contains '1234ABC'. Below the search bar, it says 'You have 3 violations:'. A light blue box displays the details of one violation:

- Violation Type:** عدم ارتداء حزام الأمان
- Fine amount:** 50000.00
- Plate Number:** 1234ABC
- City:** ريف دمشق
- street:** alhamra street
- Description:** -
- Date:** 2025-12-06 18:48:10

At the bottom of this box is a yellow 'Submit Appeal' button.

Submit appeal

Violation Details

Submit an Appeal

Reason for Appeal:

Describe why you are objecting

Send Appeal

View appeal states

Violation Type: تجاوز السرعة

Fine amount: 150.00

Plate Number: 1234ABC

City: دمشق

Street: شارع عدنان المالكي

Description: تجاوز السرعة

Date: 2025-02-10 14:30:00

Appeal Status: pending

Reason: oiluytr

Decision Note: -

Submitted: 2025-12-06T13:39:38.000000Z

Violation Type: تجاوز السرعة

Admin interface:

Login interface :

Not secure 172.20.10.8:8000/admin/login

Summarise

Chat

Admin Login

Email

Password

Login

Dashboard interface :

Admin Dashboard

Dashboard

Police Accounts

VIOLATIONS MANAGEMENT

Violation Types

Logout

Users Status Chart

Active Inactive

Last 5 Added Users

#	Name	Email	Phone	Role	Status
8	Yousuf	yousufPolice@app.com	-	Police Manager	Inactive

Total Users

6

Active Users

3

Inactive Users

3

The screenshot shows the Admin Dashboard interface. On the left is a blue sidebar with navigation links: Dashboard, Police Accounts, and Violations Management (with a sub-link for Violation Types). A Logout button is at the bottom of the sidebar. The main content area has a header with a browser address bar and a 'Summarise' button. Below the header is a summary card showing 'Inactive Users: 3' with a red background. A table titled 'Last 5 Added Users' lists the following data:

#	Name	Email	Phone	Role	Status
8	Yousuf	yousufPolice@app.com	-	Police Manager	Inactive
7	Sami Alhasan	samiPolice@app.com	-	Police Manager	Inactive
6	Mohamed Alnajar	moPolice@app.com	-	Police Officer	Inactive
4	Test User	test1@app.com	-	Police Officer	Active
1	Test User	test@example.com	-	Police Officer	Active

Manage user account:

The screenshot shows the 'Police Accounts' management page. It features a 'Create New Account' button at the top right. The main content is a table with the following data:

#	Name	Email	Phone	Role	Status	Actions
8	Yousuf	yousufPolice@app.com	—	Police Manager	Inactive	Edit, Activate, Delete
7	Sami Alhasan	samiPolice@app.com	—	Police Manager	Inactive	Edit, Activate, Delete
6	Mohamed Alnajar	moPolice@app.com	—	Police Officer	Inactive	Edit, Activate, Delete
4	Test User	test1@app.com	—	Police Officer	Active	Edit, Deactivate, Delete
1	Test User	test@example.com	—	Police Officer	Active	Edit, Deactivate, Delete

Create user account:

The screenshot shows a web browser window with the URL `172.20.10.8:8000/admin/Users/create`. The browser's address bar also shows "Not secure". The page features a blue sidebar on the left with the following menu items: "Admin Dashboard", "Dashboard", "Police Accounts" (highlighted), "VIOLATIONS MANAGEMENT", and "Violation Types". At the bottom of the sidebar is a "Logout" button. The main content area has a blue header titled "Create New Police Officer". Below this header is a form with the following fields: "Name" (text input), "Email" (text input), "Phone (Optional)" (text input), "Password" (text input), "User Role" (dropdown menu with "Police Officer" selected), and "Account Status" (dropdown menu with "Active" selected). A blue "Create User" button is positioned at the bottom of the form.

Edit user account:

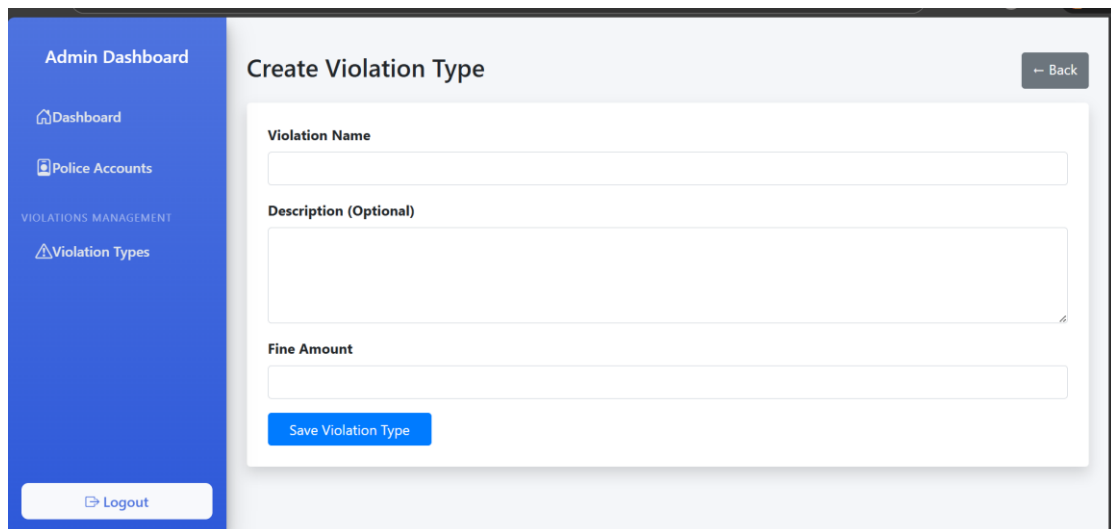
The screenshot shows a web browser window with the URL `172.20.10.8:8000/admin/Users/edit/8`. The browser's address bar also shows "Not secure". The page features a blue sidebar on the left with the following menu items: "Admin Dashboard", "Dashboard", "Police Accounts" (highlighted), "VIOLATIONS MANAGEMENT", and "Violation Types". At the bottom of the sidebar is a "Logout" button. The main content area has a light blue header titled "Edit User". Below this header is a "Back to Users" button. The form contains the following fields: "Name" (text input with "yousuf"), "Email" (text input with "yousufPolice@app.com"), "Phone" (text input with "9 digits"), and "Account Status" (dropdown menu with "Active" selected). A blue "Save Changes" button is located at the bottom of the form.

View violation type



#	Name	Description	Fine Amount	Created At
1	تجاوز السرعة		150.00	2025-11-20
2	اصطفاف خاطئ		200000.00	2025-12-03
3	القيادة بدون رخصة أو انتهاء صلاحيتها.		100000.00	2025-12-06
4	عدم ارتداء حزام الأمان		50000.00	2025-12-06
5	مخالفة إشارات المرور		150000.00	2025-12-06
6	استخدام الهاتف أثناء القيادة		180000.00	2025-12-06
7	test	...	150000.00	2025-12-06

Create violation type



Admin Dashboard

Dashboard

Police Accounts

VIOLATIONS MANAGEMENT

Violation Types

Logout

Create Violation Type

Back

Violation Name

Description (Optional)

Fine Amount

Save Violation Type

4.updated test case :

RTM Last Version:

Req-ID	Use Case	Analysis	System Design	Detailed Design	Coding	Test Case(s)
RE-FR-01	UC-1	AM-AN	User Management Service	—	Code	UC1-TC01, UC1-TC02
RE-FR-02	UC-1	AM-AN	User Management Service	—	Code	UC1-TC03
RE-FR-03	UC-2	AM-AN	User Management Service	—	Code	UC2-TC01, UC2-TC02
RE-FR-04	UC-3	AM-AN	Violation + AI Services	—	Code	UC3-TC01
RE-FR-05	UC-4	AM-AN	Violation Management	—	Code	UC4-TC01, UC4-TC02
RE-FR-06	UC-3	AM-AN	Plate Recognition Service	—	Code	UC3-TC02, UC3-TC03
RE-FR-07	UC-3	AM-AN	Violation Management	—	Code	UC3-TC04
RE-FR-08	UC-3	AM-AN	Violation Management	—	Code	UC3-TC05
RE-FR-09	UC-5	BM-AN	Violation Management	—	Code	UC5-TC01
RE-FR-10	UC-5	BM-AN	Violation Management	—	Code	UC5-TC02
RE-FR-11	UC-11	BM-AN	Violation Management	—	Code	UC11-TC01
RE-FR-12	UC-12	BM-AN	Appeal Management	—	Code	UC12-TC01
RE-FR-13	UC-12	BM-AN	Appeal Management	—	Code	UC12-TC02

RE-FR-14	UC-1	AM-AN	User Management Service	—	Code	UC1-TC01
RE-FR-15	UC-1	AM-AN	User Management Service	—	Code	UC1-TC03
RE-FR-16	UC-2	AM-AN	User Management Service	—	Code	UC2-TC01
RE-FR-17	UC-6	BM-AN	User Management Service	—	Code	UC6-TC01
RE-FR-18	UC-7	BM-AN	User Management Service	—	Code	UC7-TC01
RE-FR-19	UC-7	BM-AN	User Management Service	—	Code	UC7-TC02
RE-FR-20	UC-8	BM-AN	User Management Service	—	Code	UC8-TC01
RE-FR-21	UC-6	BM-AN	User Management Service	—	Code	UC6-TC02
RE-FR-22	UC-10	DM-AN	Appeal Management	—	Code	UC10-TC01
RE-FR-23	UC-10	DM-AN	Appeal Management	—	Code	UC10-TC02
RE-FR-24	UC-10	DM-AN	Appeal Management	—	Code	UC10-TC03
RE-FR-25	UC-3	FGM-AN	AI – Speech to Text	—	Code	UC3-TC06
RE-FR-26	UC-3	FGM-AN	AI – Plate Recognition	—	Code	UC3-TC07
RE-FR-27	UC-3	FGM-AN	AI – Violation Generation	—	Code	UC3-TC08
RE-FR-28	UC-3	FGM-AN	AI – Classification	—	Code	UC3-TC09

Chapter 6 Conclusion

Introduction

As a result of this project, we have developed a fully functional **Traffic Police Assistant System** designed to support traffic police officers in efficiently recording, managing, and reviewing traffic violations. The system aims to replace traditional manual processes with a smart, technology-driven solution that improves accuracy, speed, and reliability in field operations.

The system enables officers to capture vehicle images and automatically extract critical information such as license plate numbers, vehicle model, and color using **Optical Character Recognition (OCR)** technologies. In addition, officers can record verbal descriptions of violations through **Speech-to-Text (STT)** functionality, allowing violation details to be transcribed automatically and used to populate violation records. This significantly reduces manual data entry and enhances usability in real-world traffic scenarios.

To ensure scalability, maintainability, and clean separation of concerns, the backend was developed following modern software engineering principles and a service-oriented architecture. The system integrates a RESTful API built with **Laravel**, a relational database for structured data management, and **RabbitMQ** to handle asynchronous AI tasks such as OCR and STT processing. This architecture allows the system to efficiently manage heavy workloads and supports future extensions.

Throughout the development of this project, our main objectives were to:

Improve the efficiency and accuracy of traffic violation recording.

Reduce manual effort for traffic police officers in the field.

Provide a secure, user-friendly, and reliable system.

Ensure scalability and extensibility through clean, modular design and modern backend practices.

References

[Laravel Documentation \(Latest Version\)](#)

[PHP Official Documentation](#)

[Flutter Documentation](#)

[RabbitMQ Documentation](#)

[RESTful API Design Best Practices](#)

[GitHub Documentation](#)

Appendices

GitHub Repositories

Backend:

https://github.com/almasri-maryam/Triffic_Police_assistant.git

Frontend (Flutter App):

https://github.com/simo75-t/grad_project.git